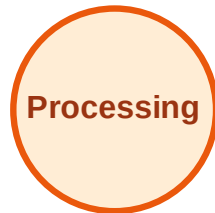


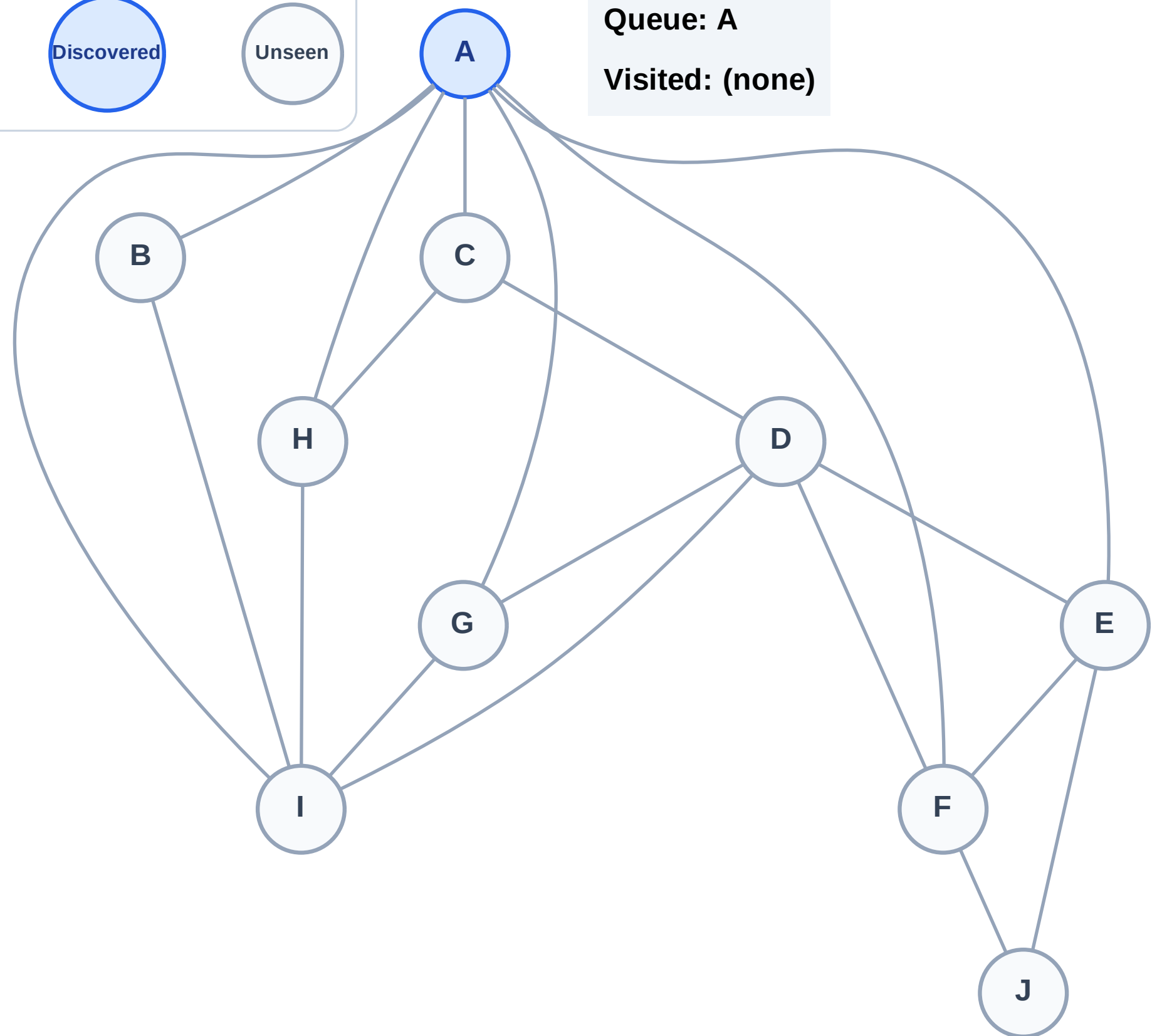
BFS Traversal

Step 1: Start at A

Legend



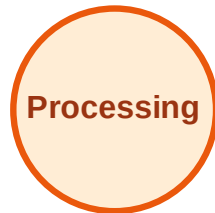
Queue: A
Visited: (none)



BFS Traversal

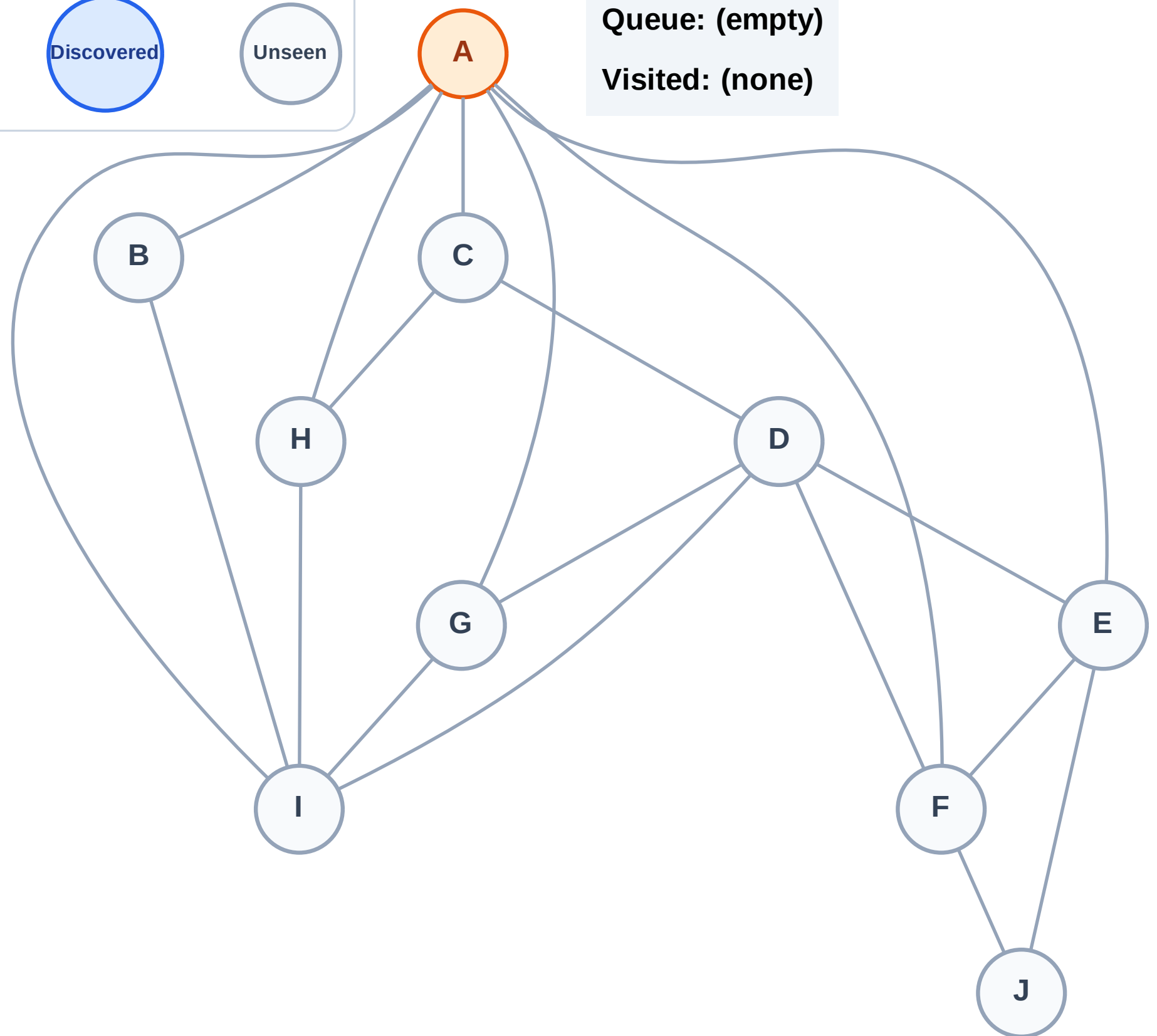
Step 2: Process node A

Legend



Queue: (empty)

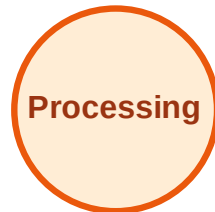
Visited: (none)



BFS Traversal

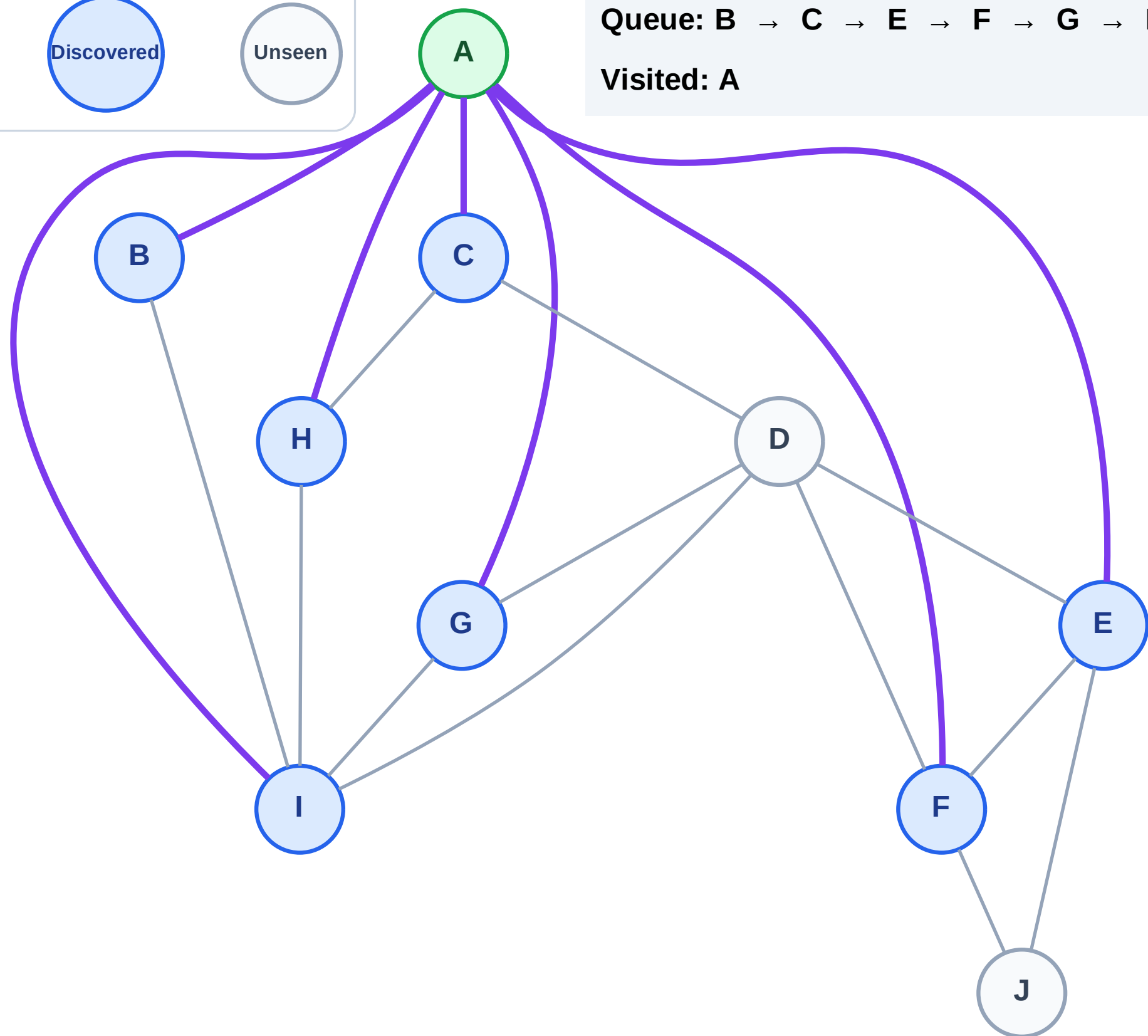
Step 3: Discover B, C, E, F, G, H, I from A

Legend



Queue: B → C → E → F → G → H → I

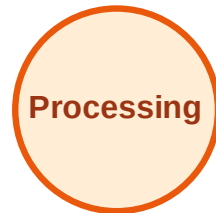
Visited: A



BFS Traversal

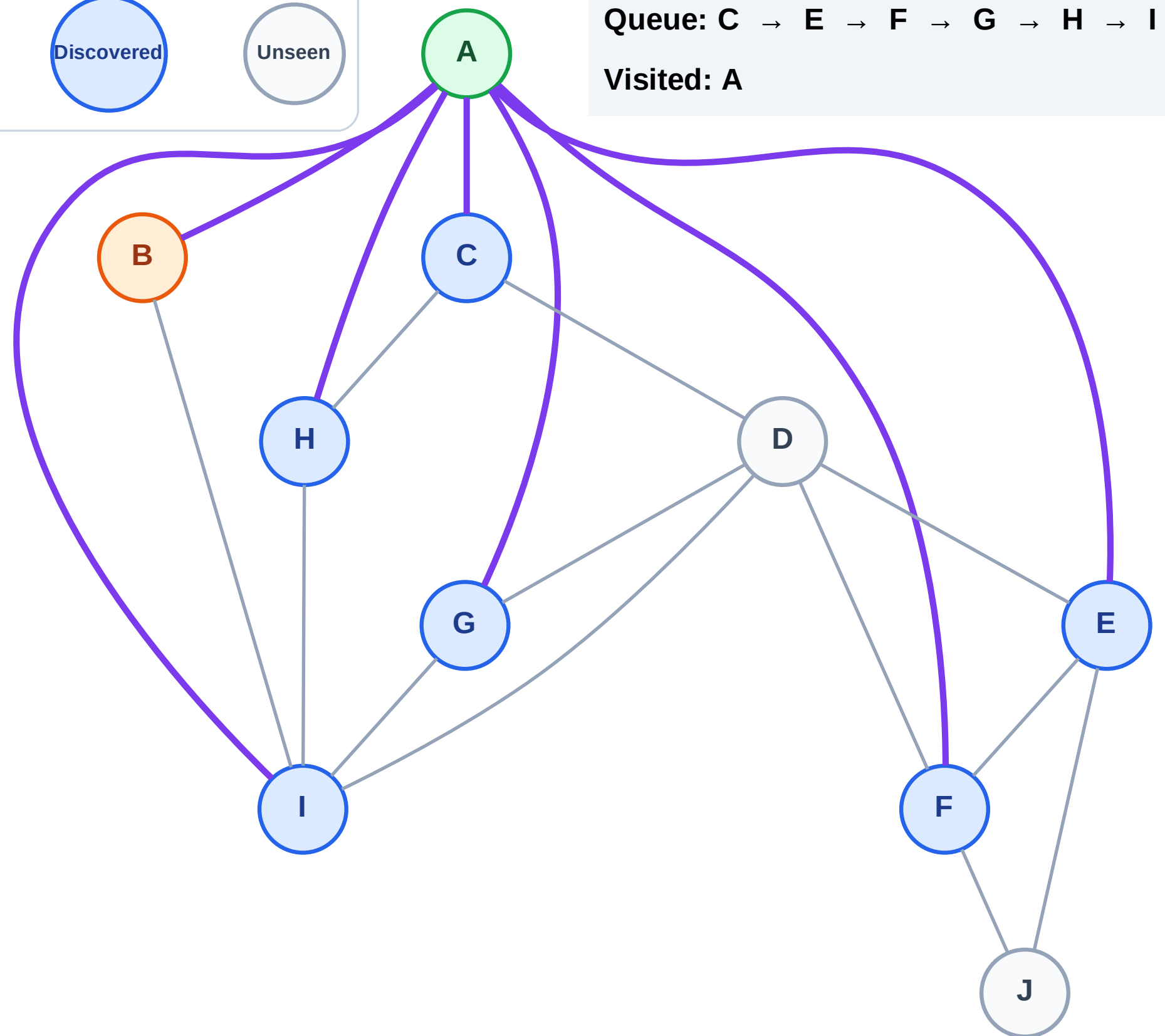
Step 4: Process node B

Legend



Queue: C → E → F → G → H → I

Visited: A



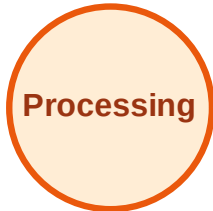
BFS Traversal

Step 5: No new neighbors from B

Legend



Complete



Processing



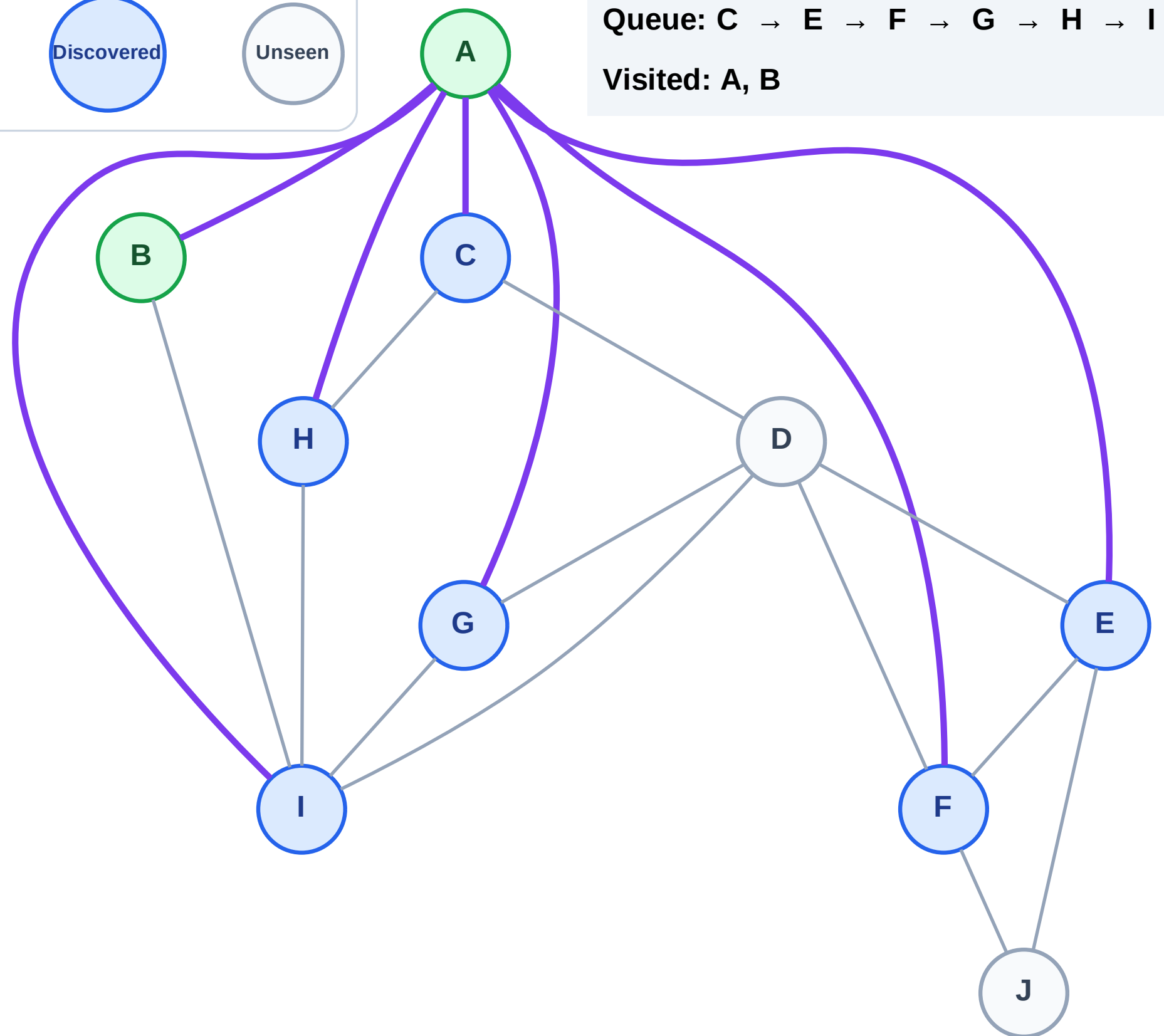
Discovered



Unseen

Queue: C → E → F → G → H → I

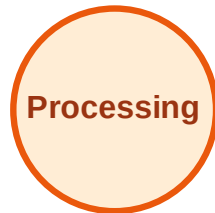
Visited: A, B



BFS Traversal

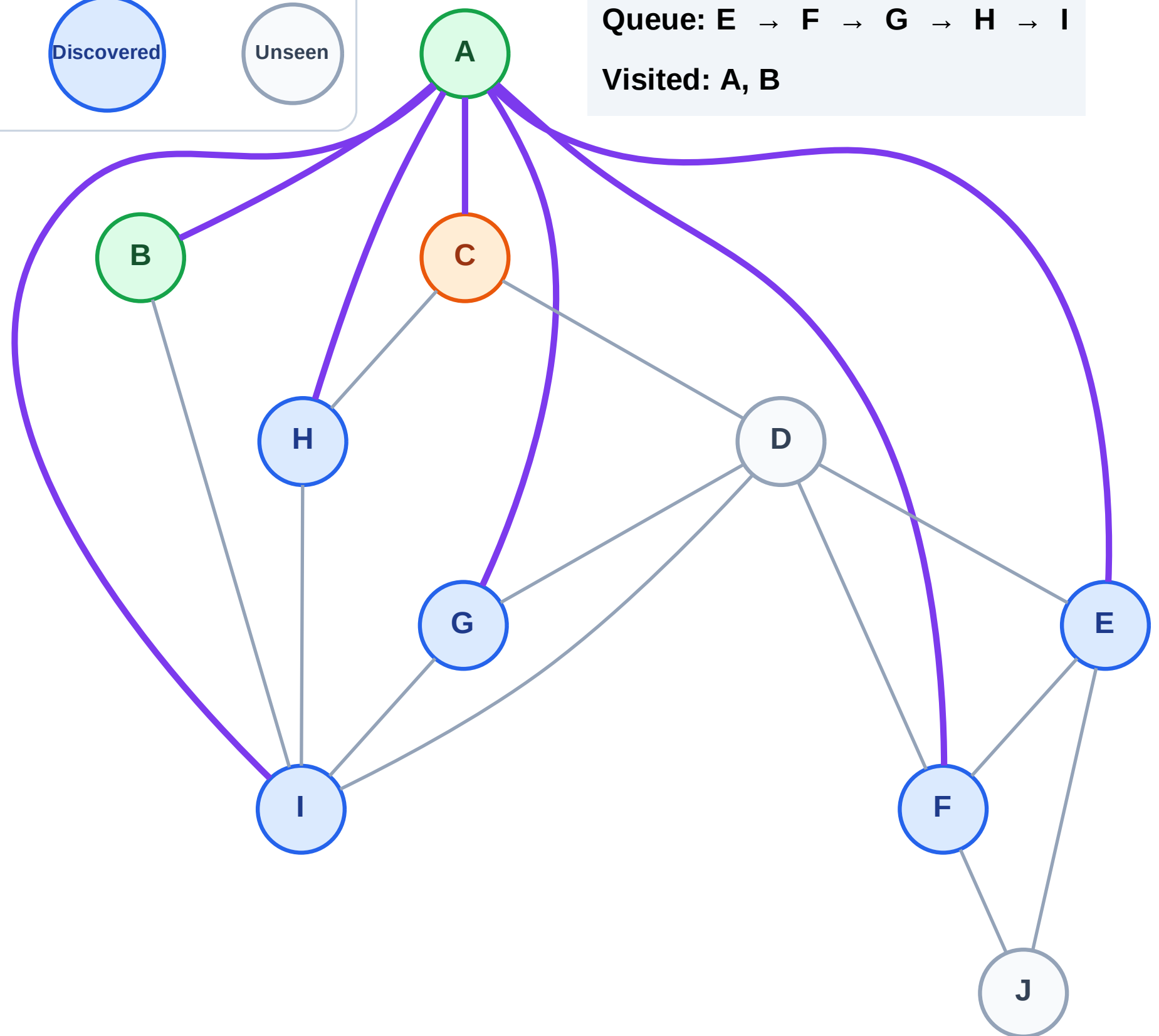
Step 6: Process node C

Legend



Queue: E → F → G → H → I

Visited: A, B



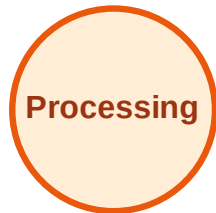
BFS Traversal

Step 7: Discover D from C

Legend



Complete



Processing



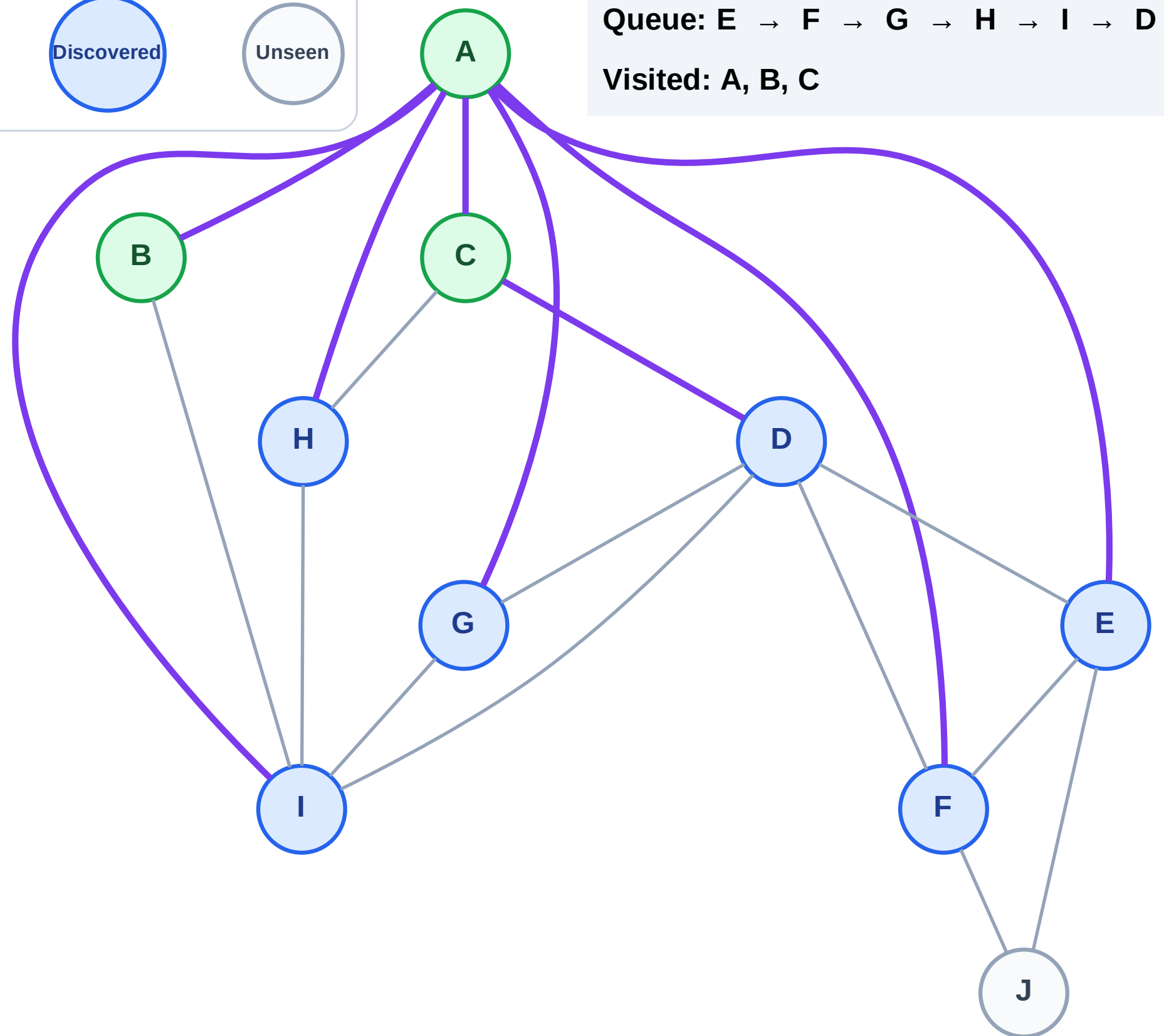
Discovered



Unseen

Queue: E → F → G → H → I → D

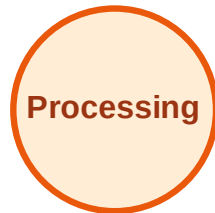
Visited: A, B, C



BFS Traversal

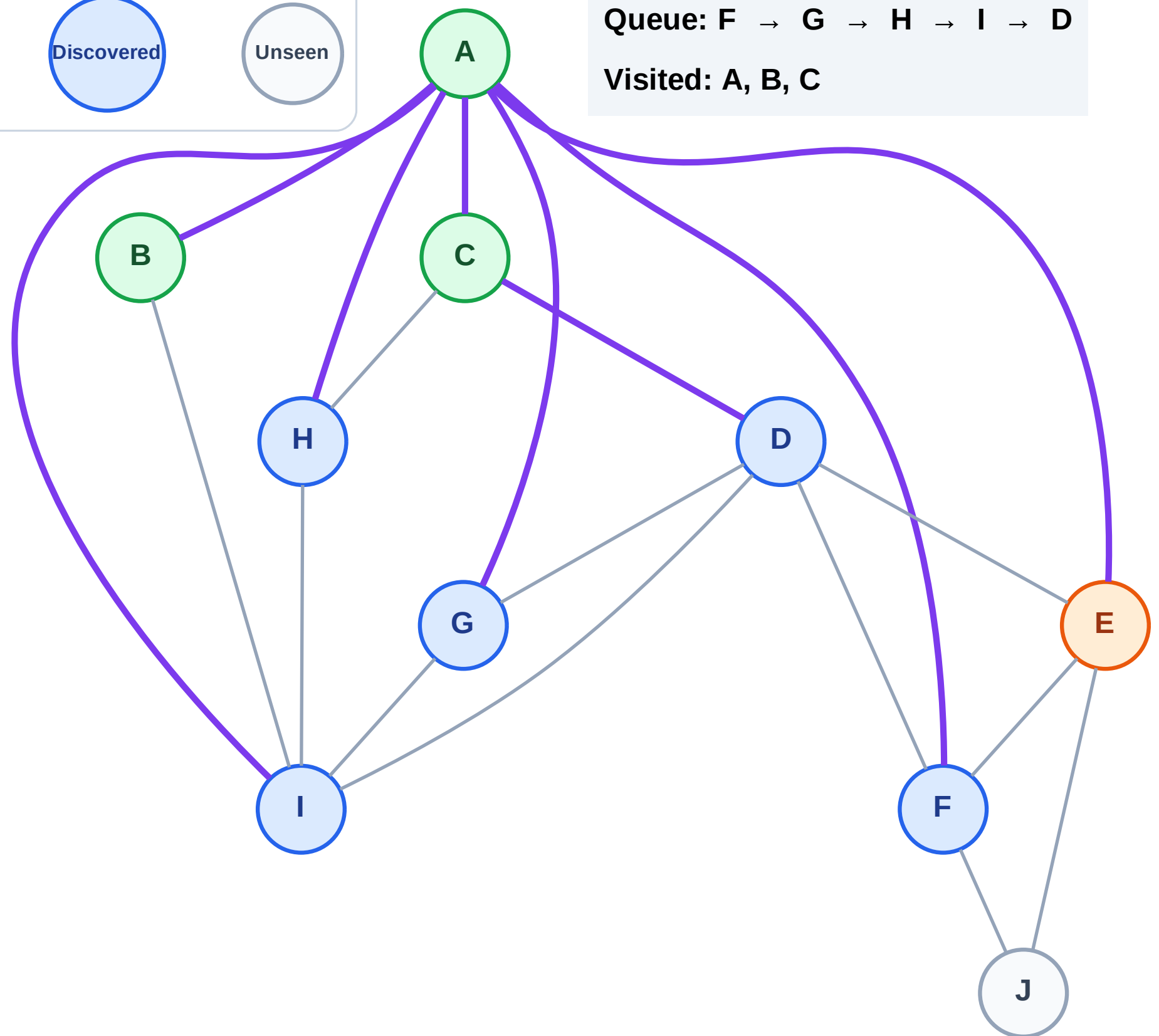
Step 8: Process node E

Legend



Queue: F → G → H → I → D

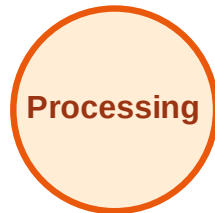
Visited: A, B, C



BFS Traversal

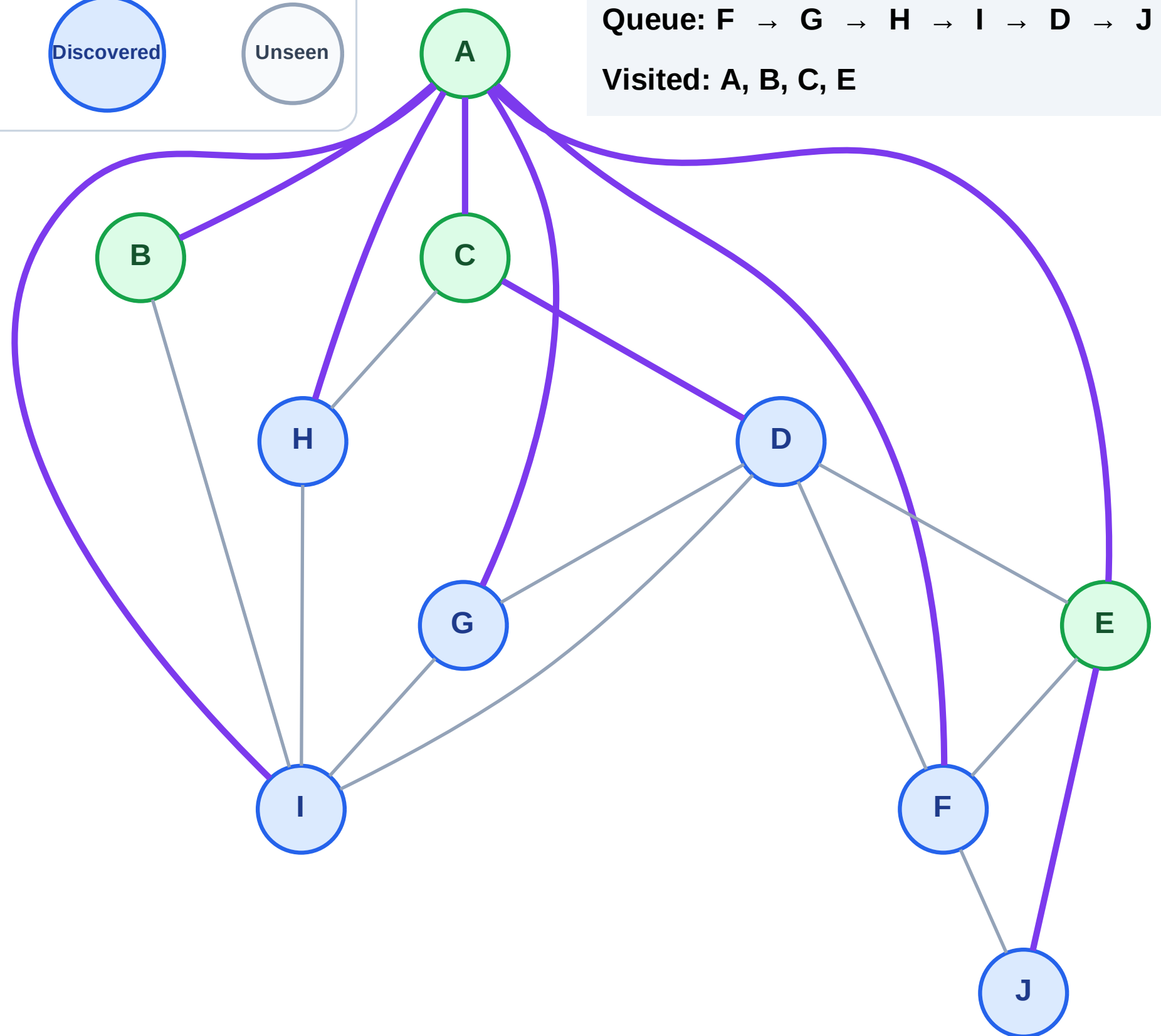
Step 9: Discover J from E

Legend



Queue: F → G → H → I → D → J

Visited: A, B, C, E



BFS Traversal

Step 10: Process node F

Legend

Complete

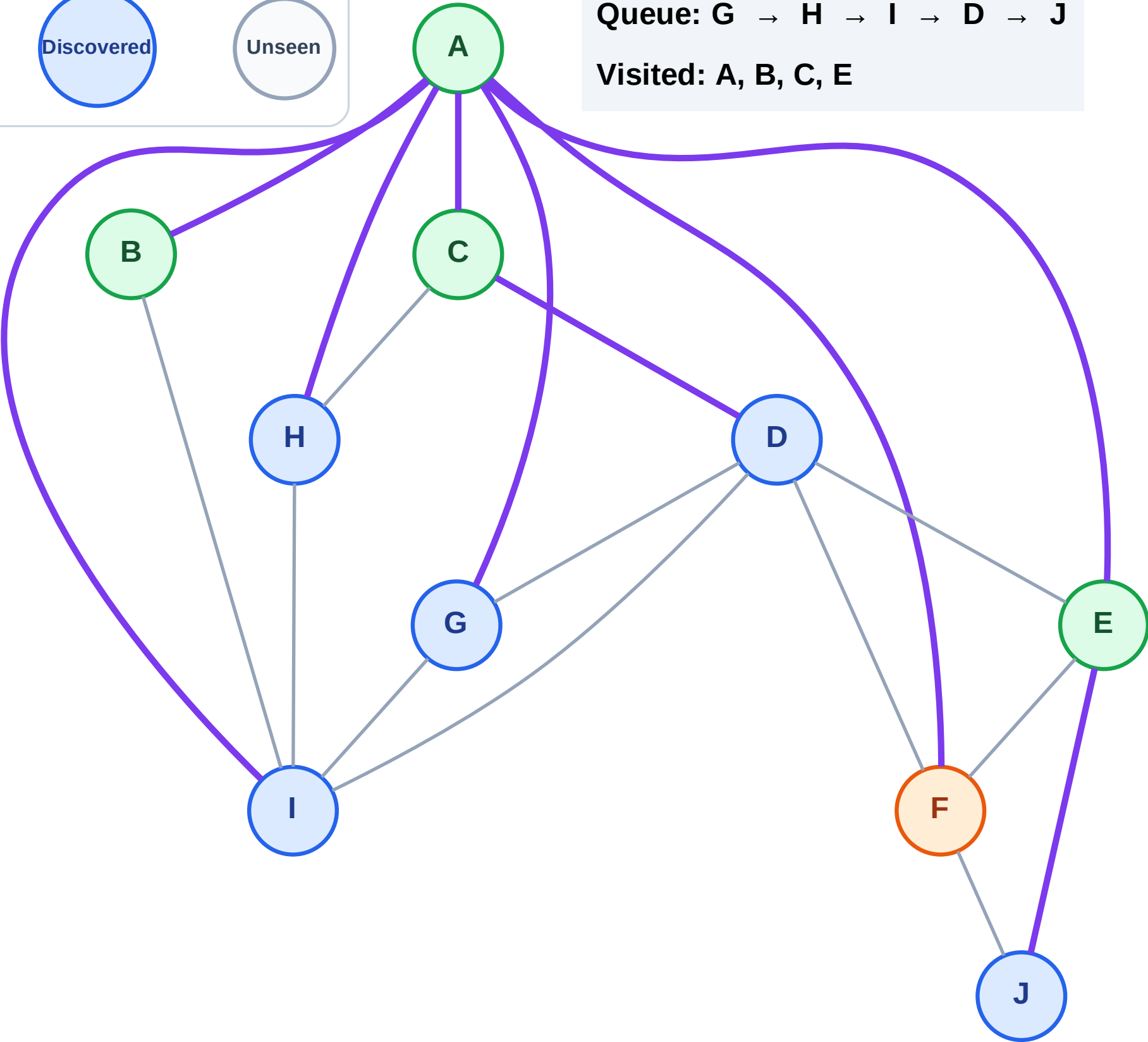
Processing

Discovered

Unseen

Queue: G → H → I → D → J

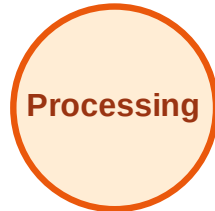
Visited: A, B, C, E



BFS Traversal

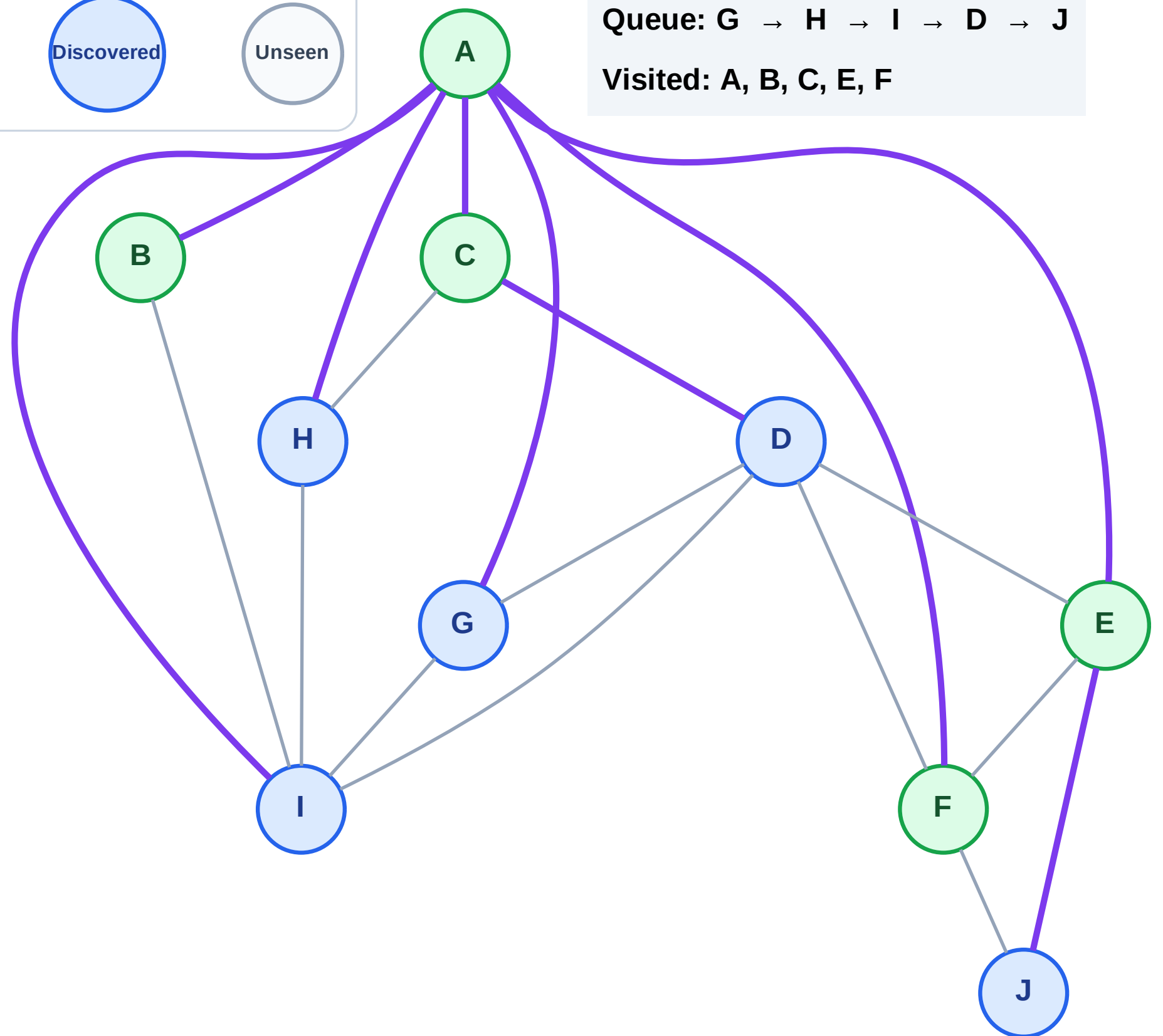
Step 11: No new neighbors from F

Legend



Queue: G → H → I → D → J

Visited: A, B, C, E, F



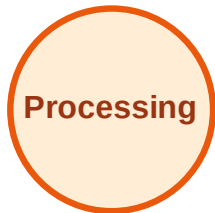
BFS Traversal

Step 12: Process node G

Legend



Complete



Processing



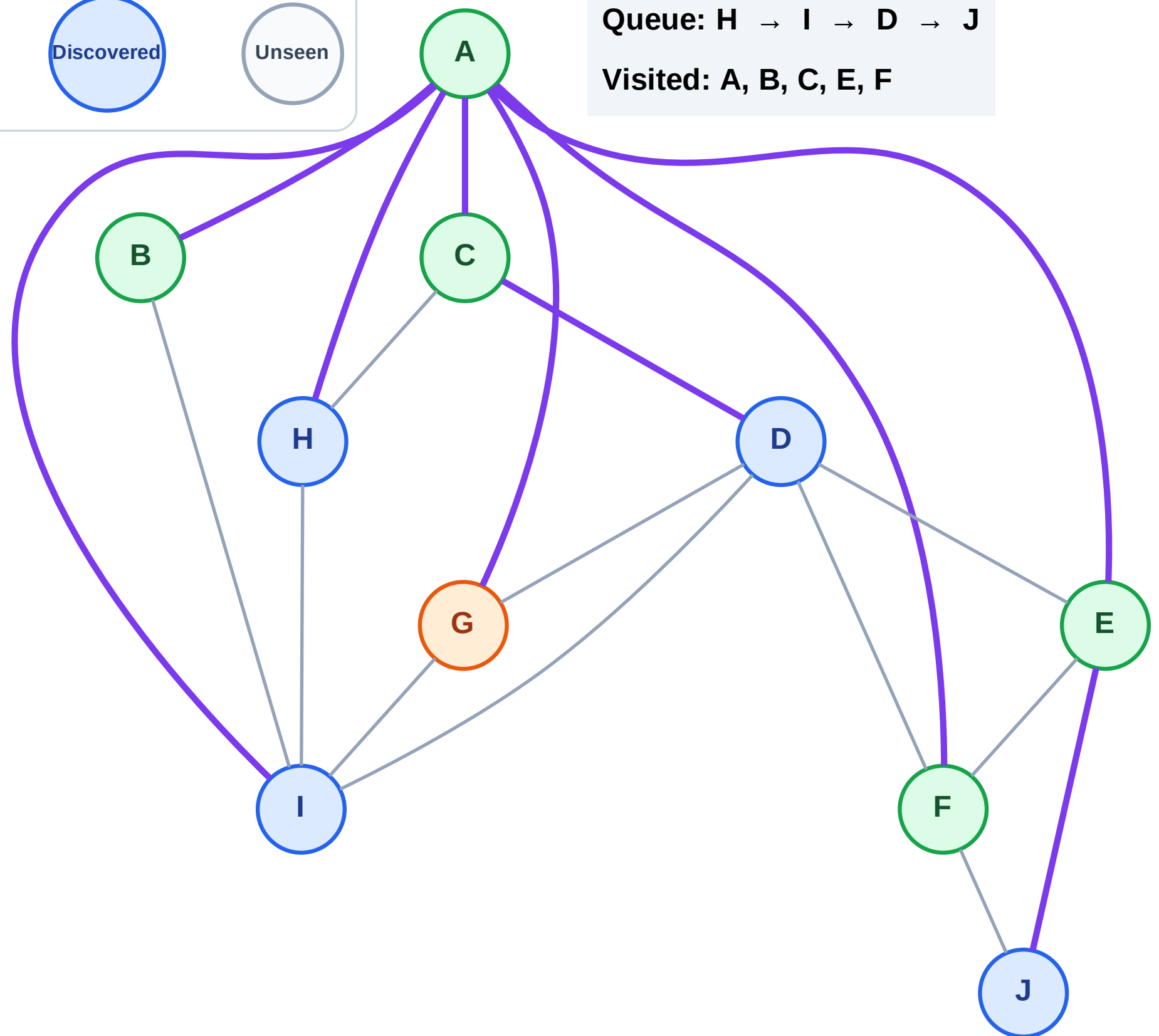
Discovered



Unseen

Queue: H → I → D → J

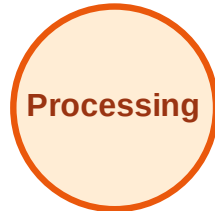
Visited: A, B, C, E, F



BFS Traversal

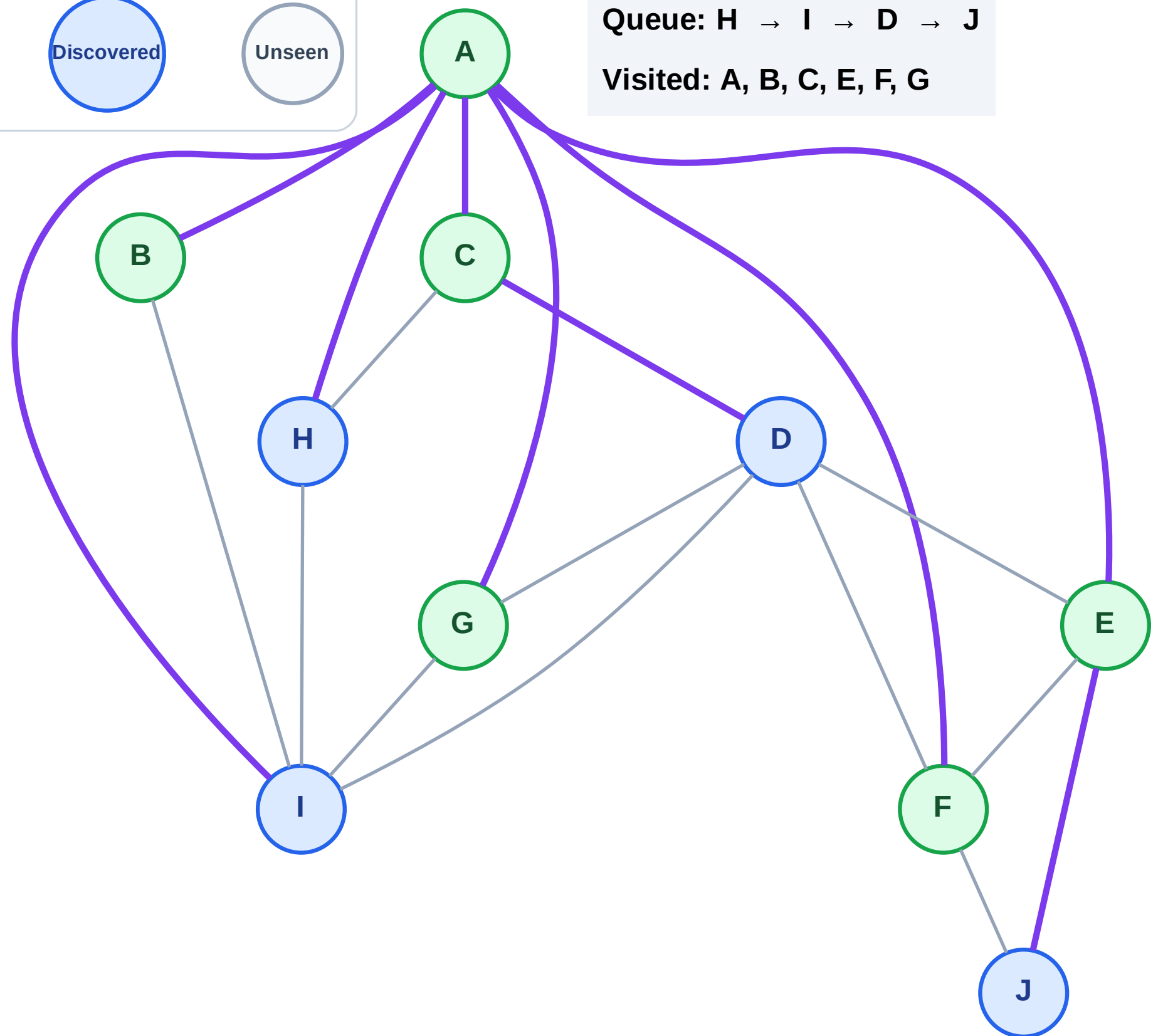
Step 13: No new neighbors from G

Legend



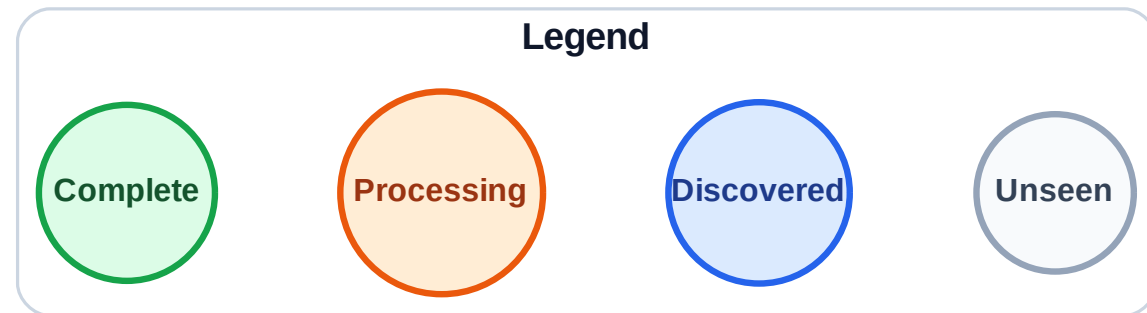
Queue: H → I → D → J

Visited: A, B, C, E, F, G



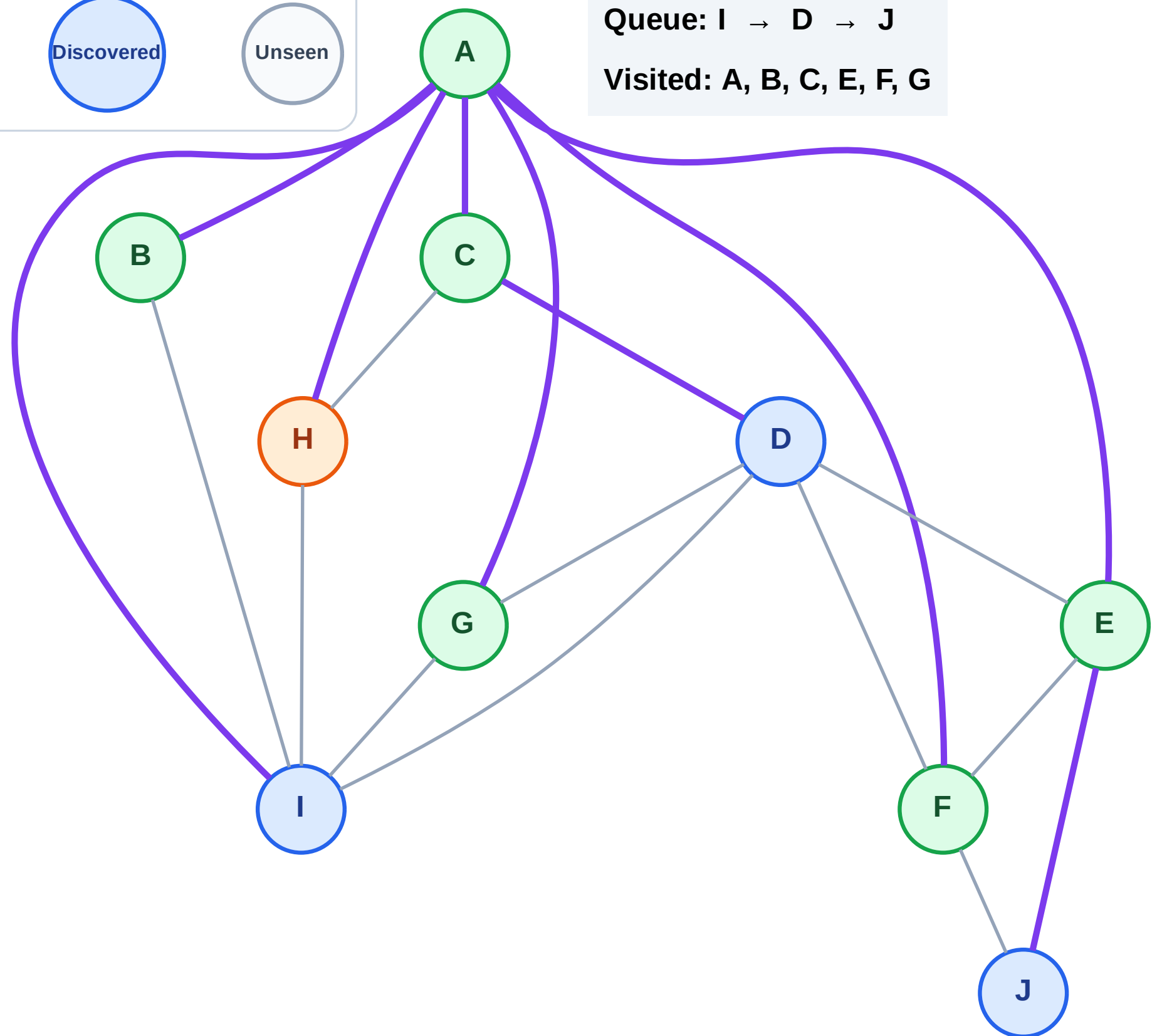
Step 14: Process node H

Step 14: Process node H



Queue: I → D → J

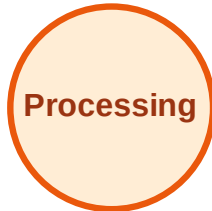
Visited: A, B, C, E, F, G



BFS Traversal

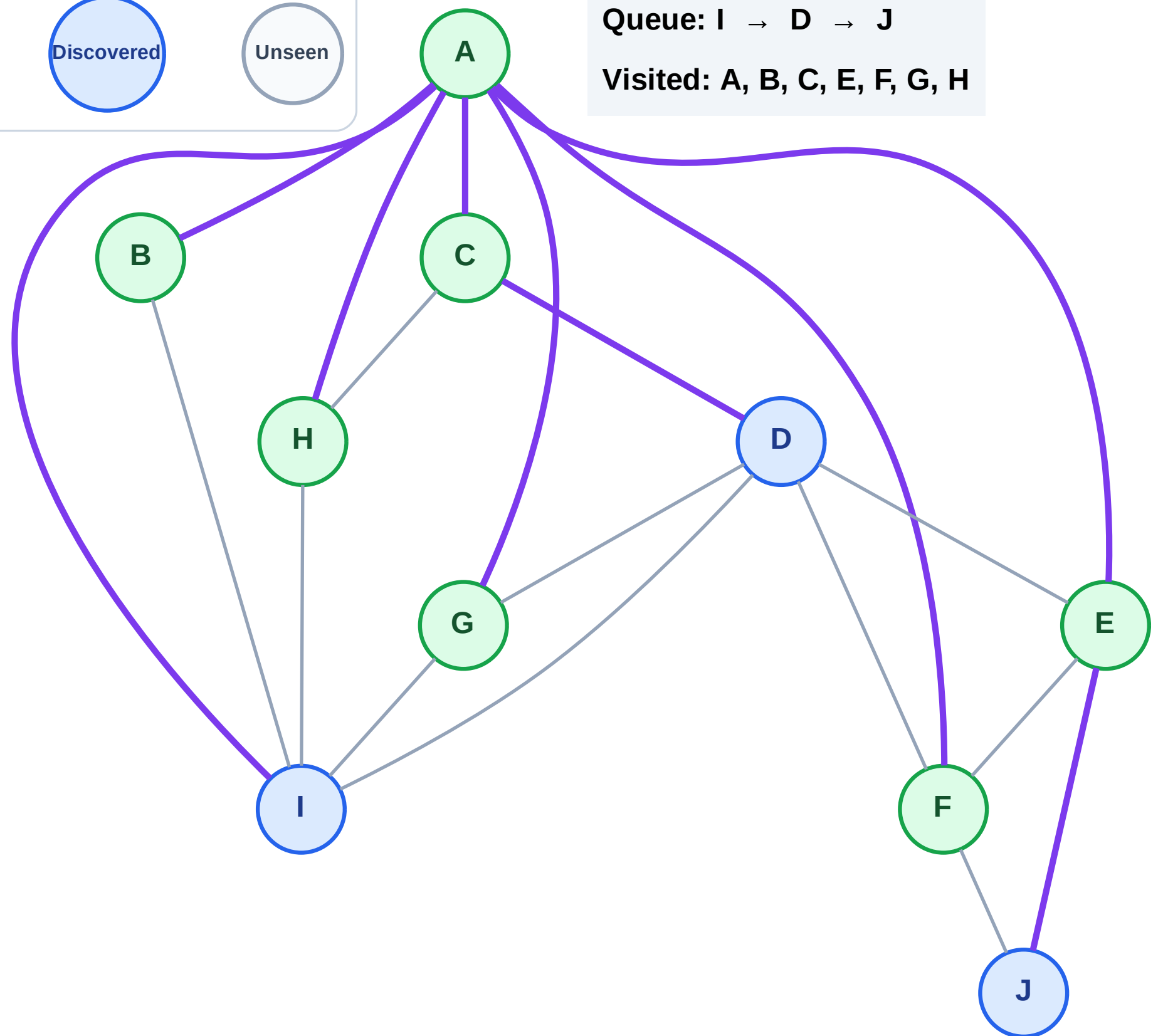
Step 15: No new neighbors from H

Legend



Queue: I → D → J

Visited: A, B, C, E, F, G, H



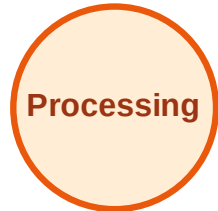
BFS Traversal

Step 16: Process node I

Legend



Complete



Processing



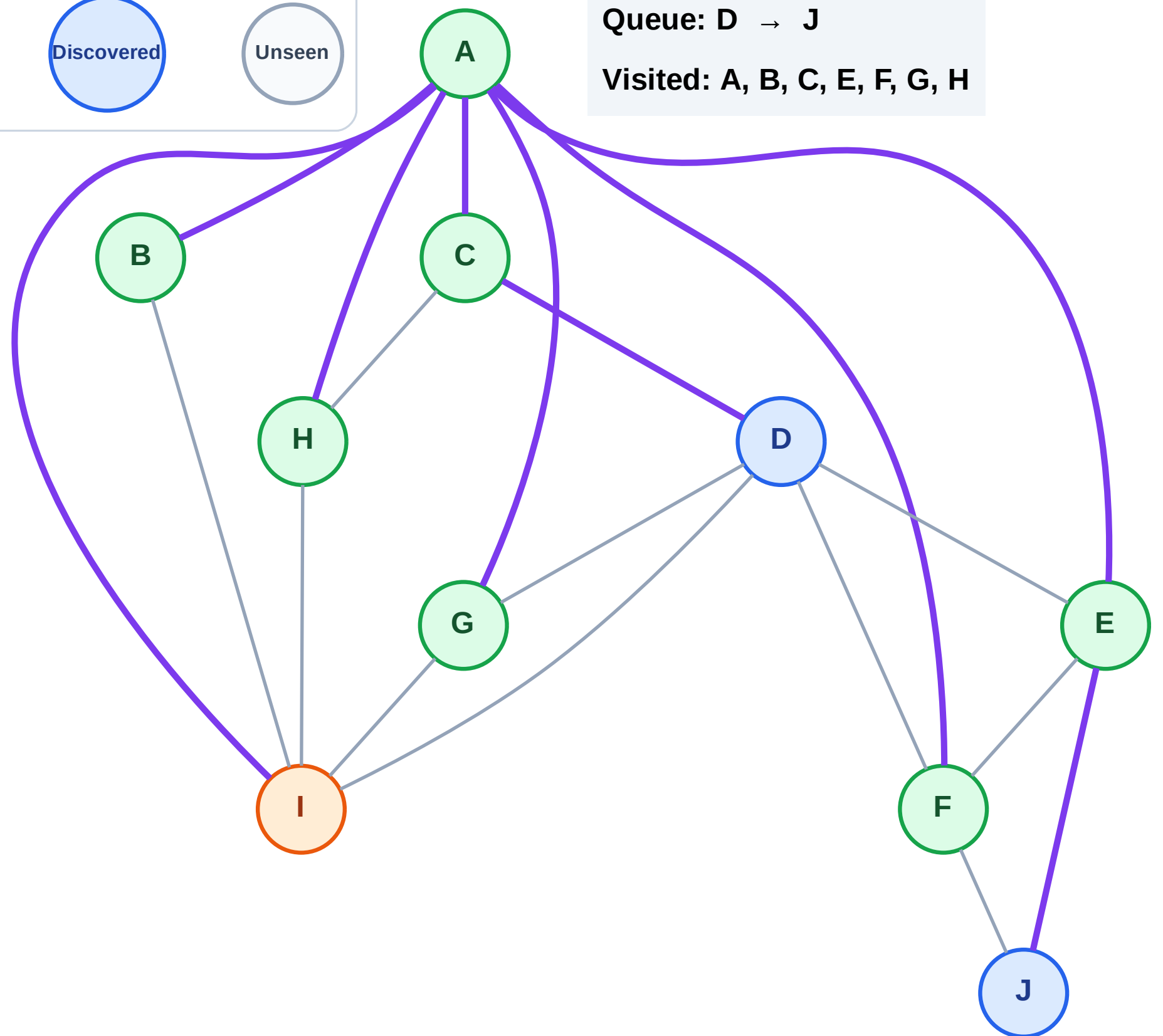
Discovered



Unseen

Queue: D → J

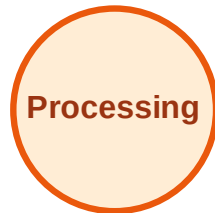
Visited: A, B, C, E, F, G, H



BFS Traversal

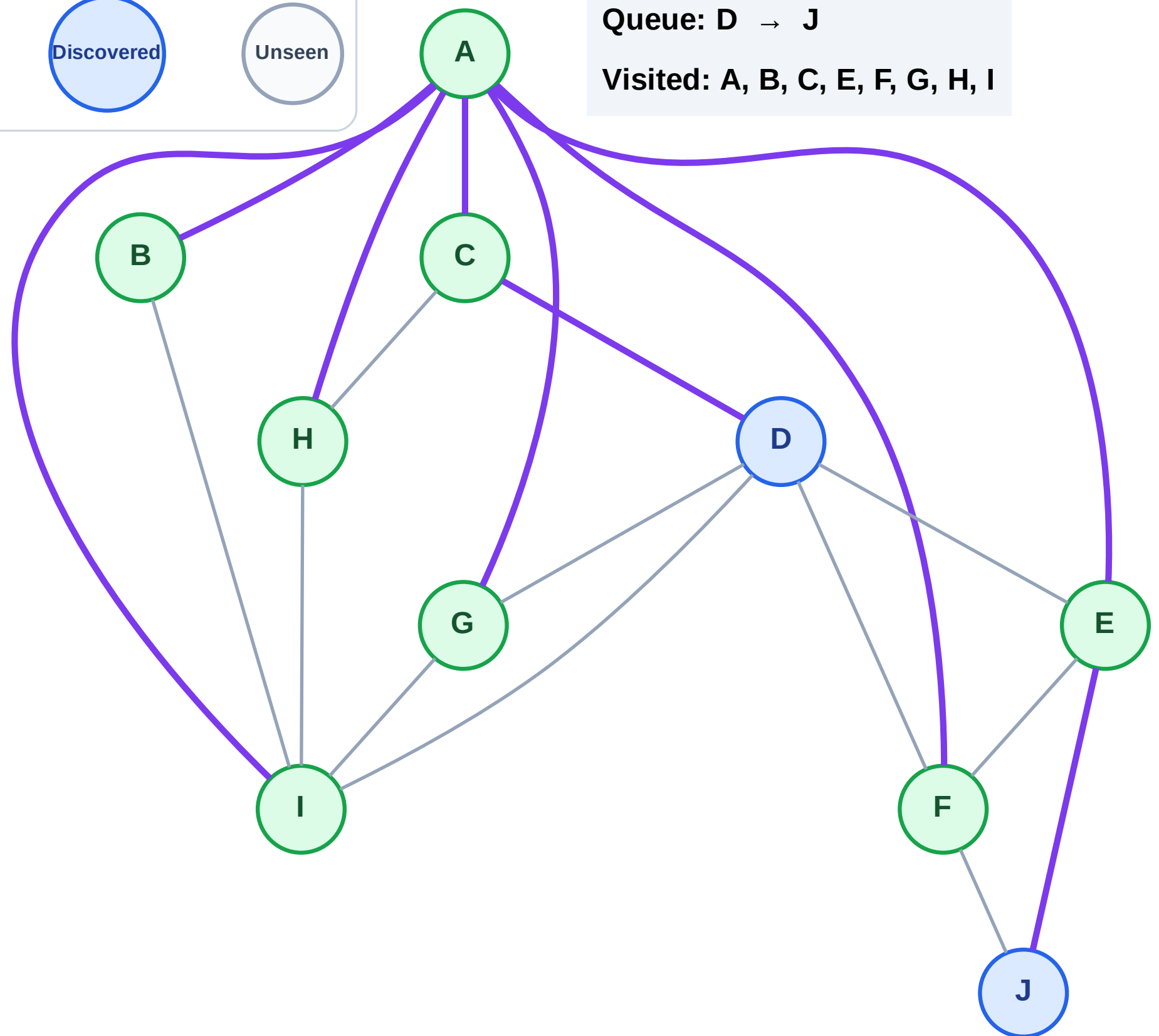
Step 17: No new neighbors from I

Legend



Queue: D → J

Visited: A, B, C, E, F, G, H, I



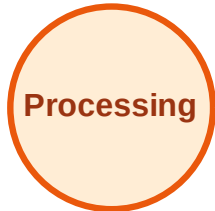
BFS Traversal

Step 18: Process node D

Legend



Complete



Processing



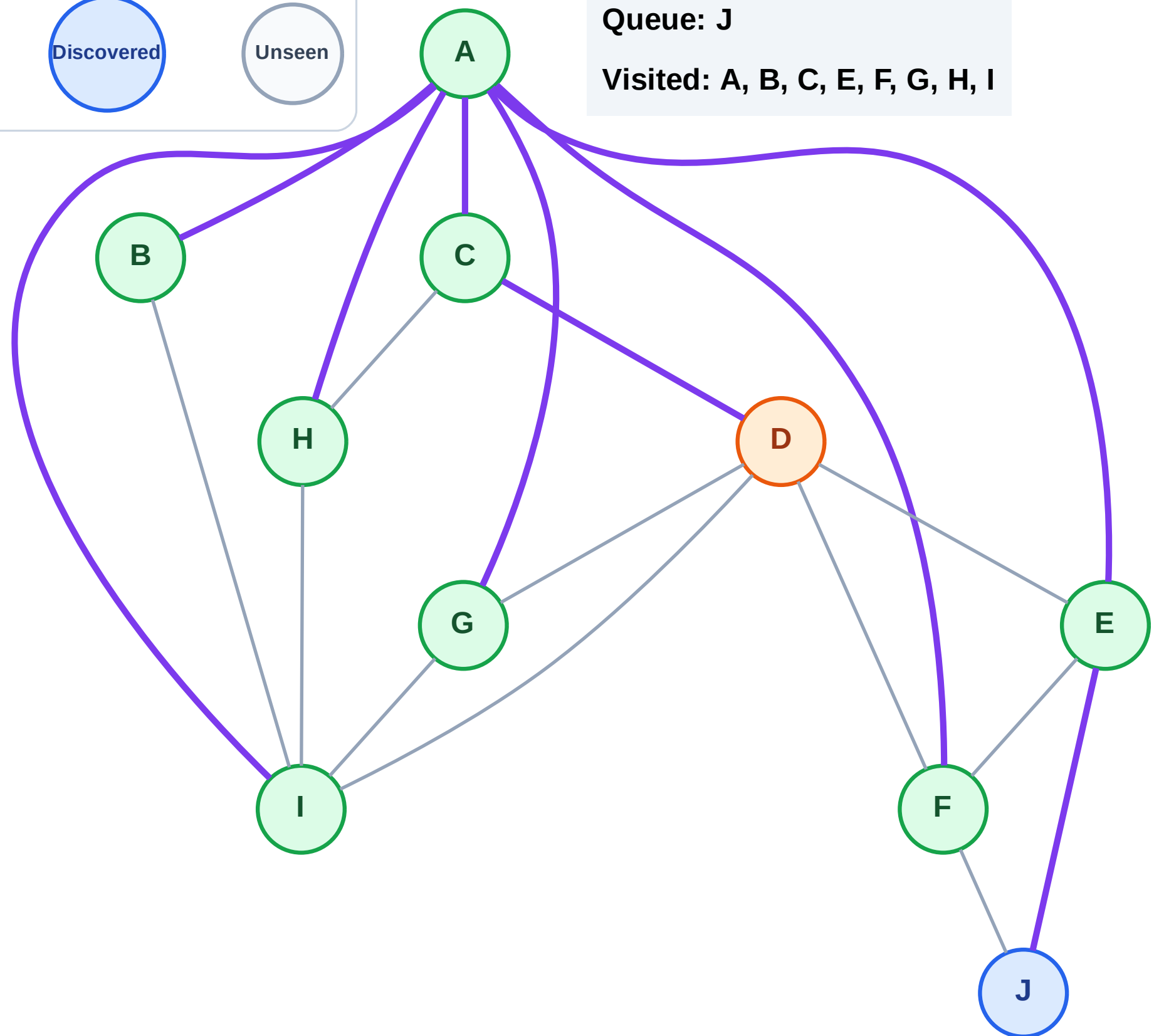
Discovered



Unseen

Queue: J

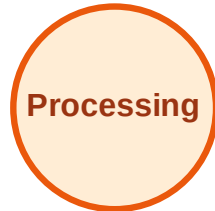
Visited: A, B, C, E, F, G, H, I



BFS Traversal

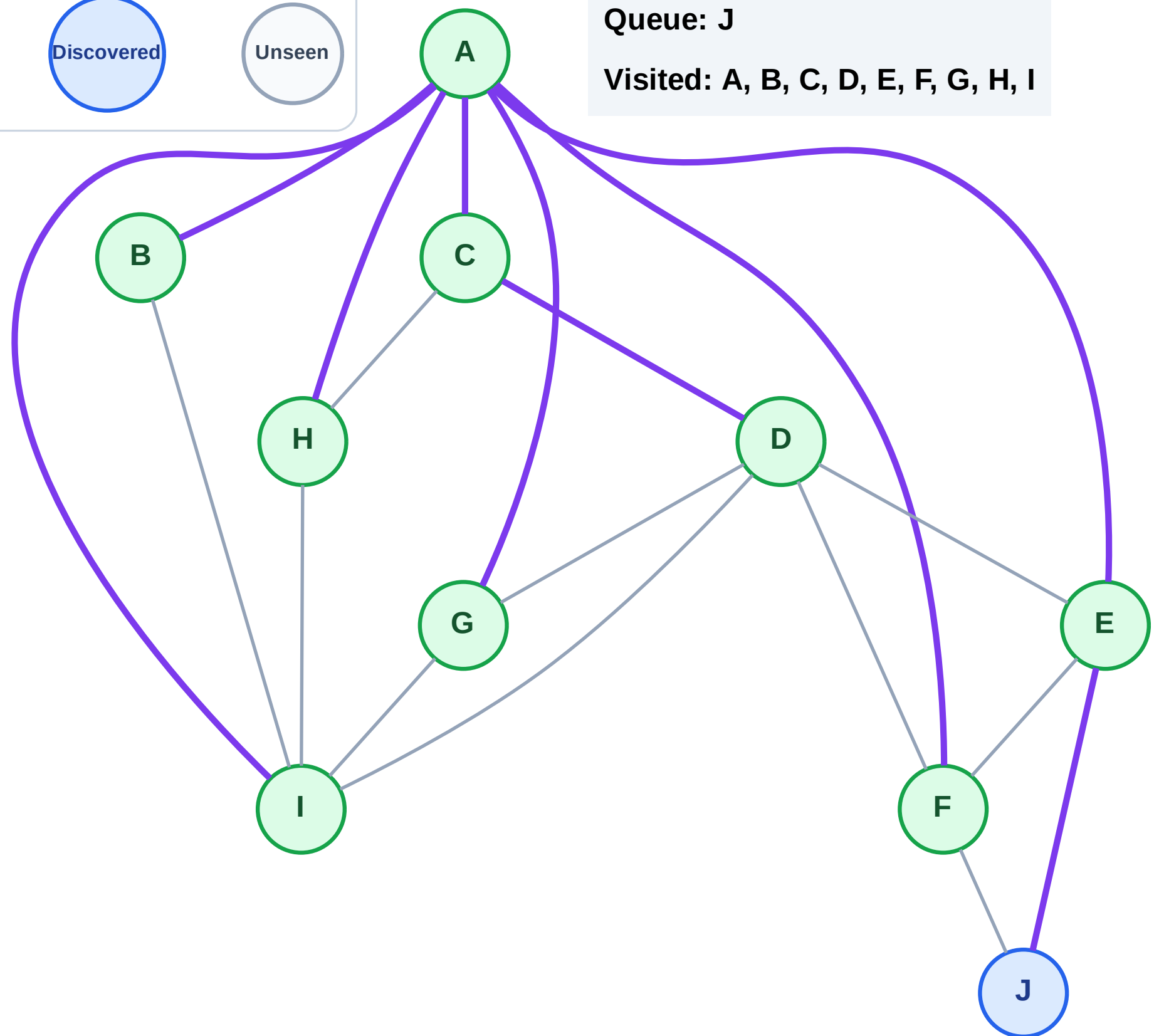
Step 19: No new neighbors from D

Legend



Queue: J

Visited: A, B, C, D, E, F, G, H, I



BFS Traversal

Step 20: Process node J

Legend

Complete

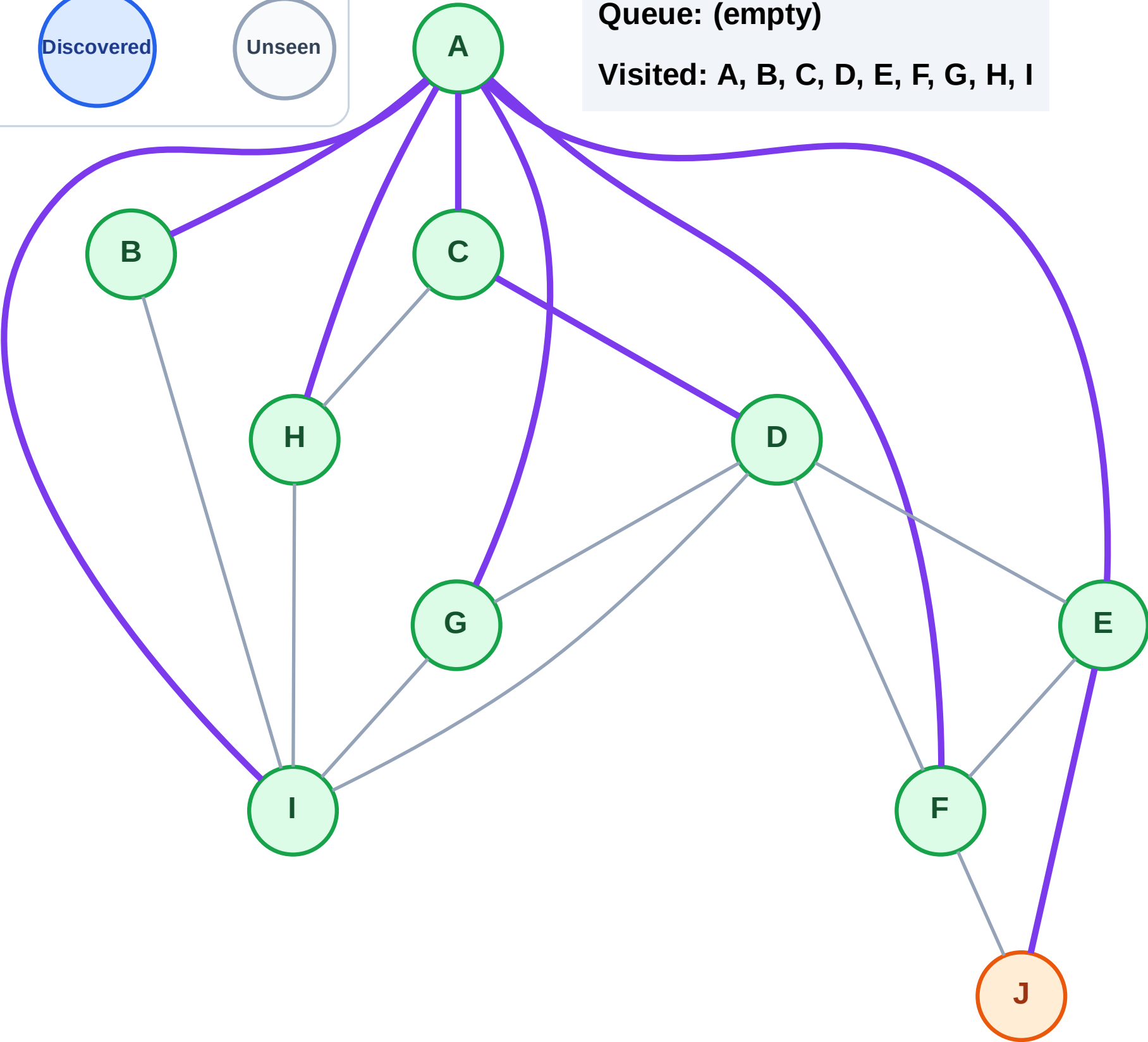
Processing

Discovered

Unseen

Queue: (empty)

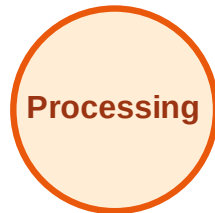
Visited: A, B, C, D, E, F, G, H, I



BFS Traversal

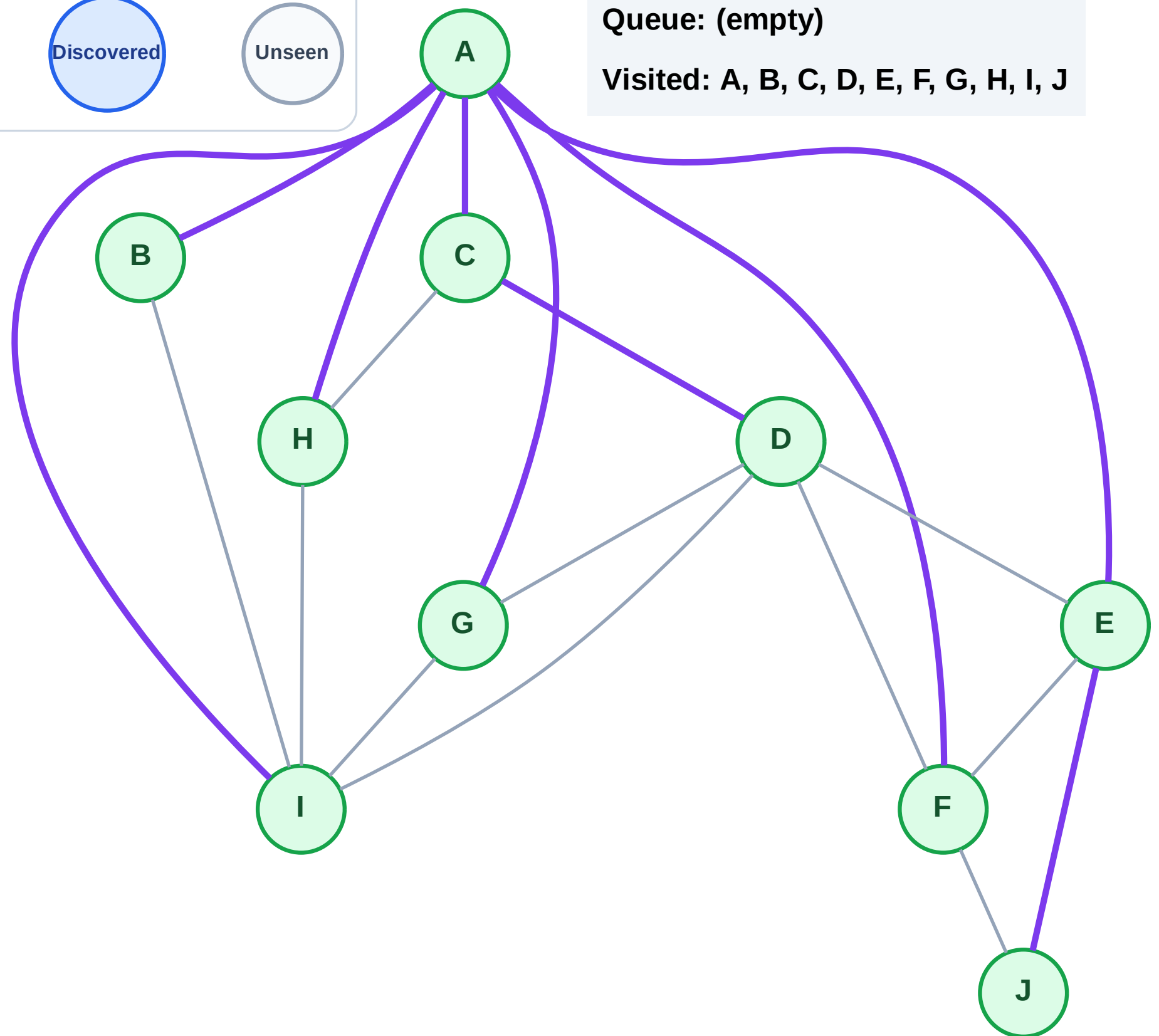
Step 21: No new neighbors from J

Legend



Queue: (empty)

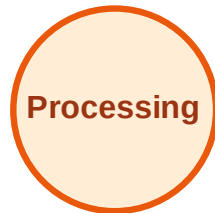
Visited: A, B, C, D, E, F, G, H, I, J



DFS Traversal

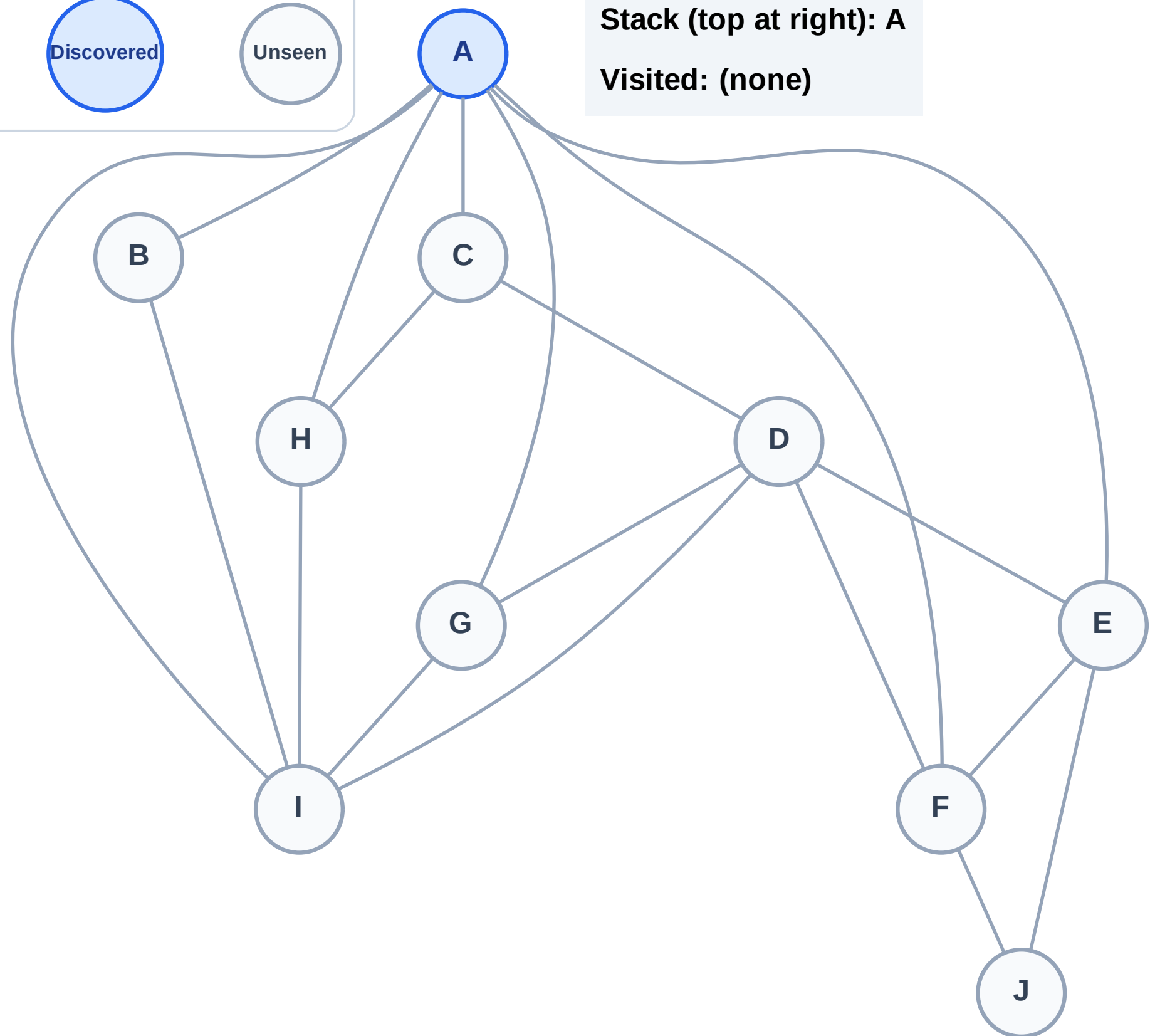
Step 1: Start at A

Legend



Stack (top at right): A

Visited: (none)



DFS Traversal

Step 2: Process node A

Legend

Complete

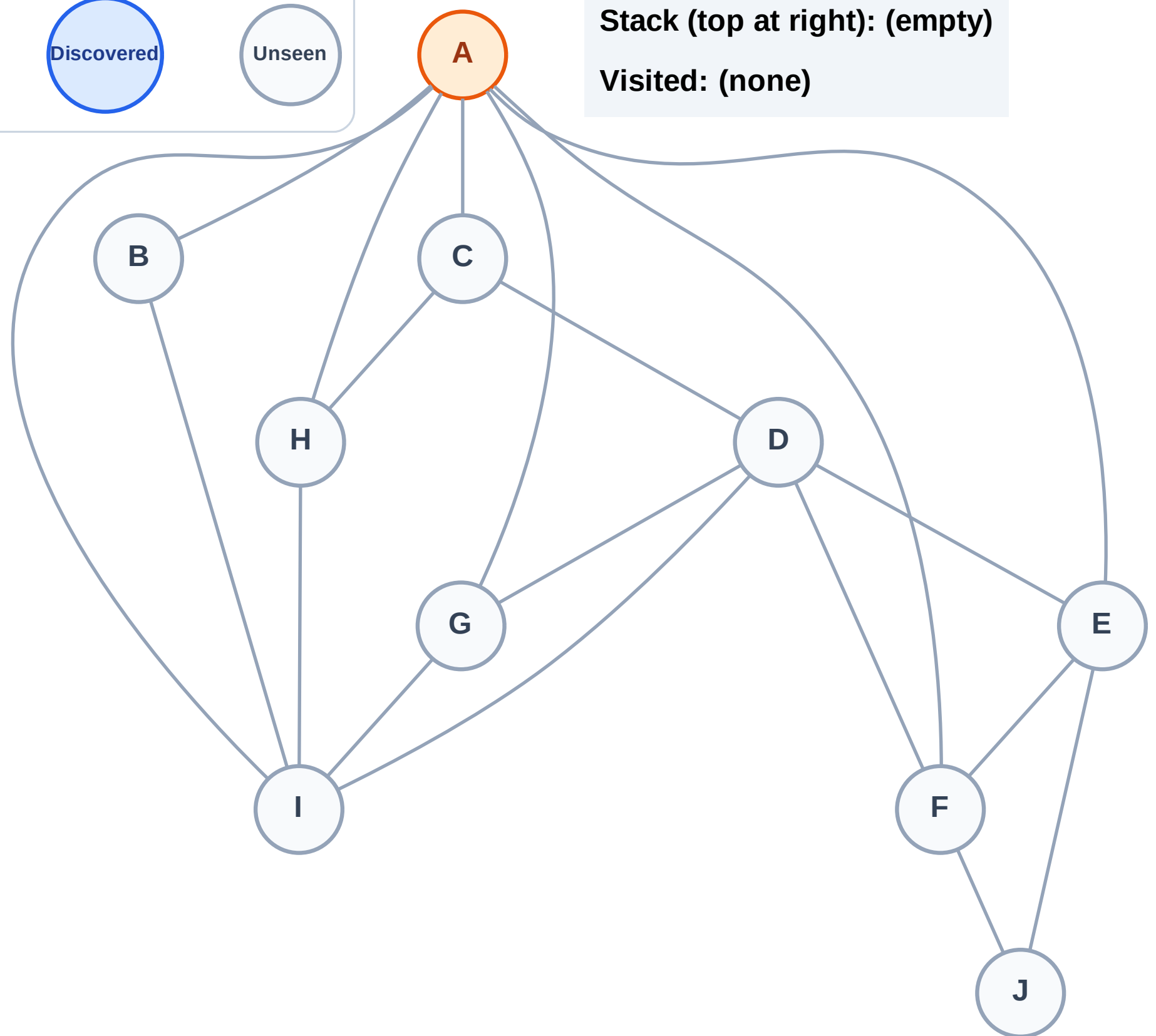
Processing

Discovered

Unseen

Stack (top at right): (empty)

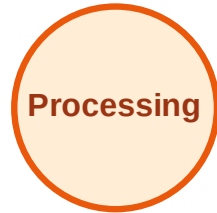
Visited: (none)



DFS Traversal

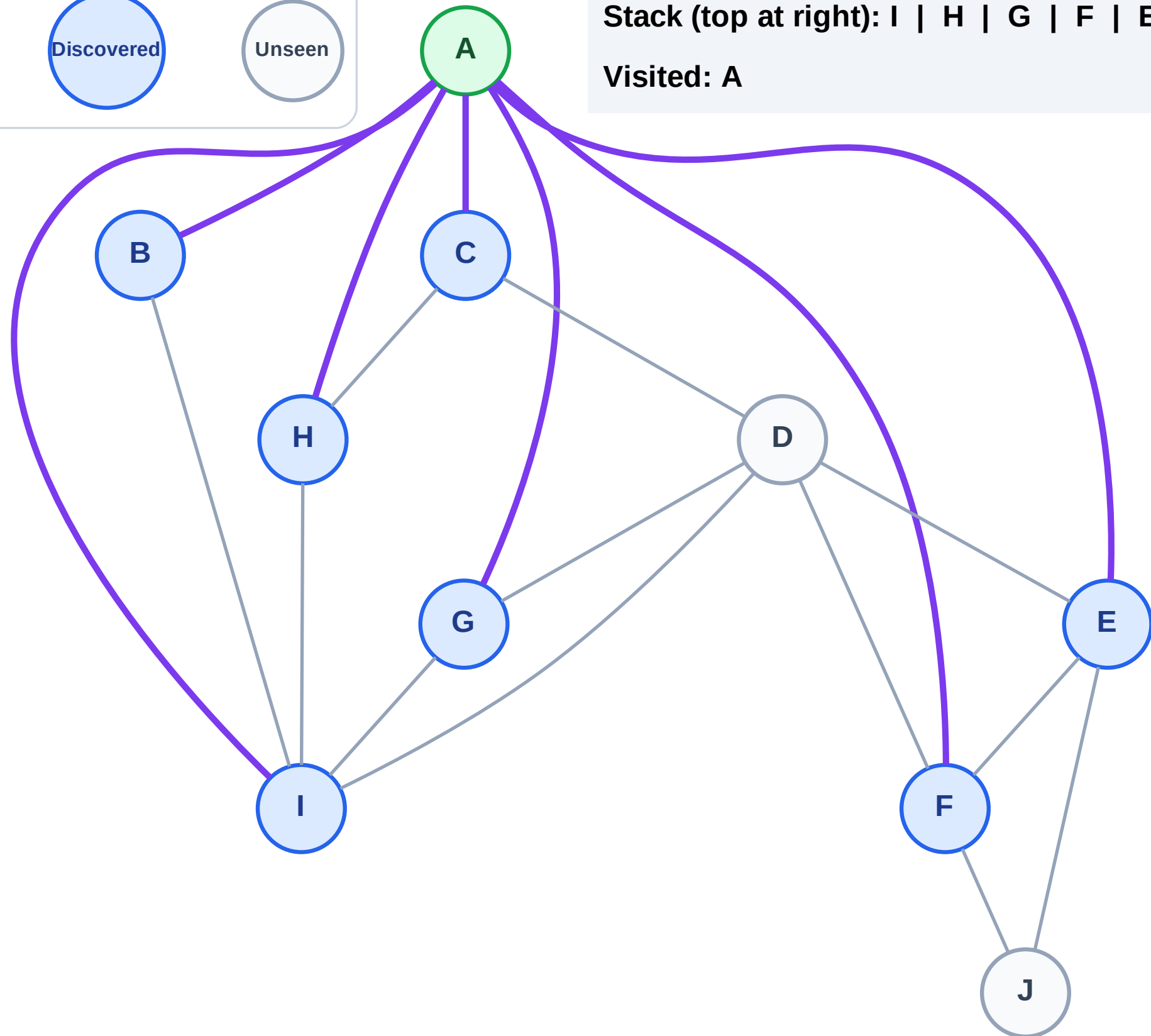
Step 3: Discover I, H, G, F, E, C, B from A

Legend



Stack (top at right): I | H | G | F | E | C | B

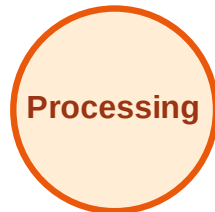
Visited: A



DFS Traversal

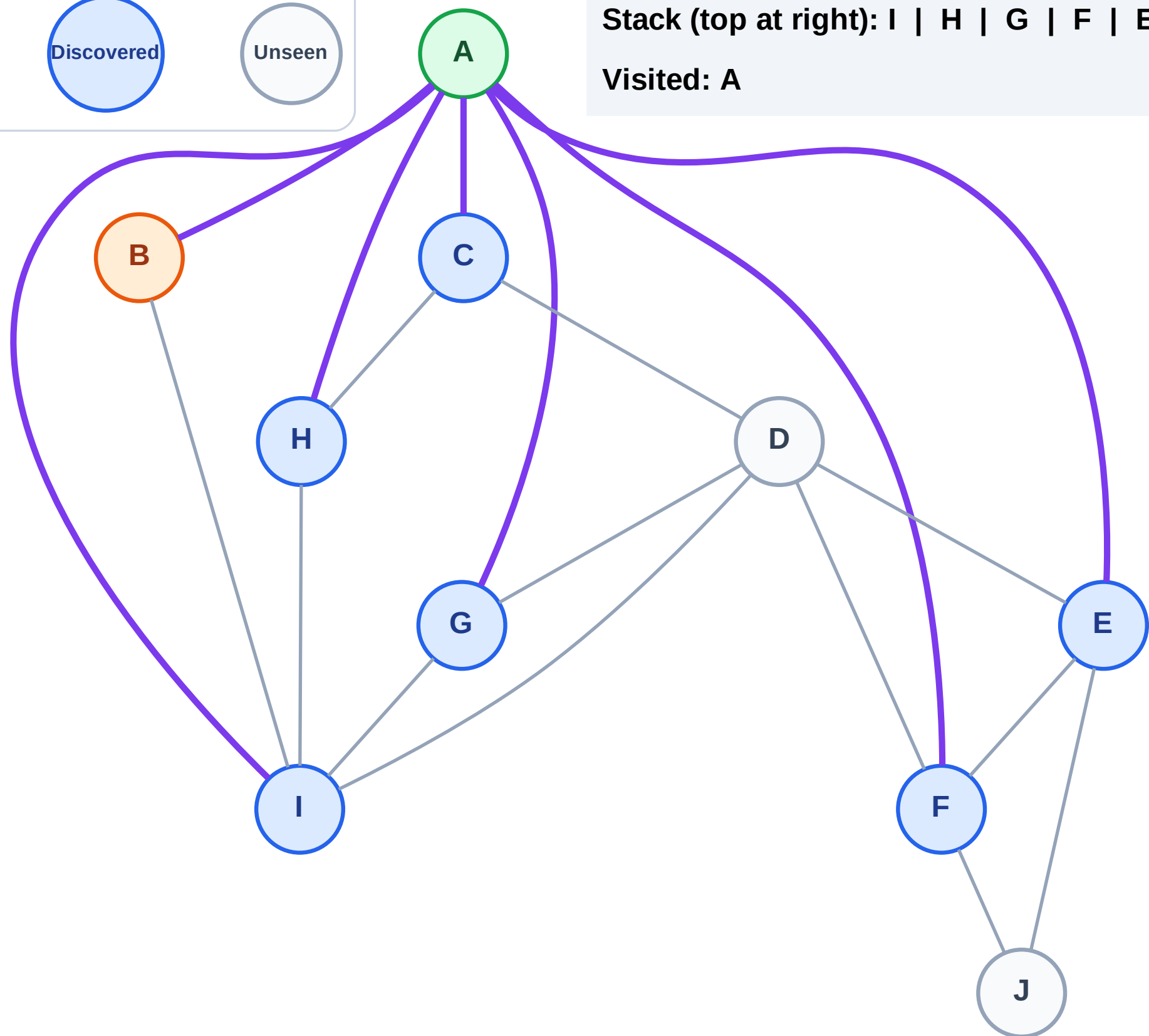
Step 4: Process node B

Legend



Stack (top at right): I | H | G | F | E | C

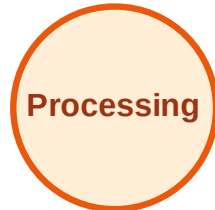
Visited: A



DFS Traversal

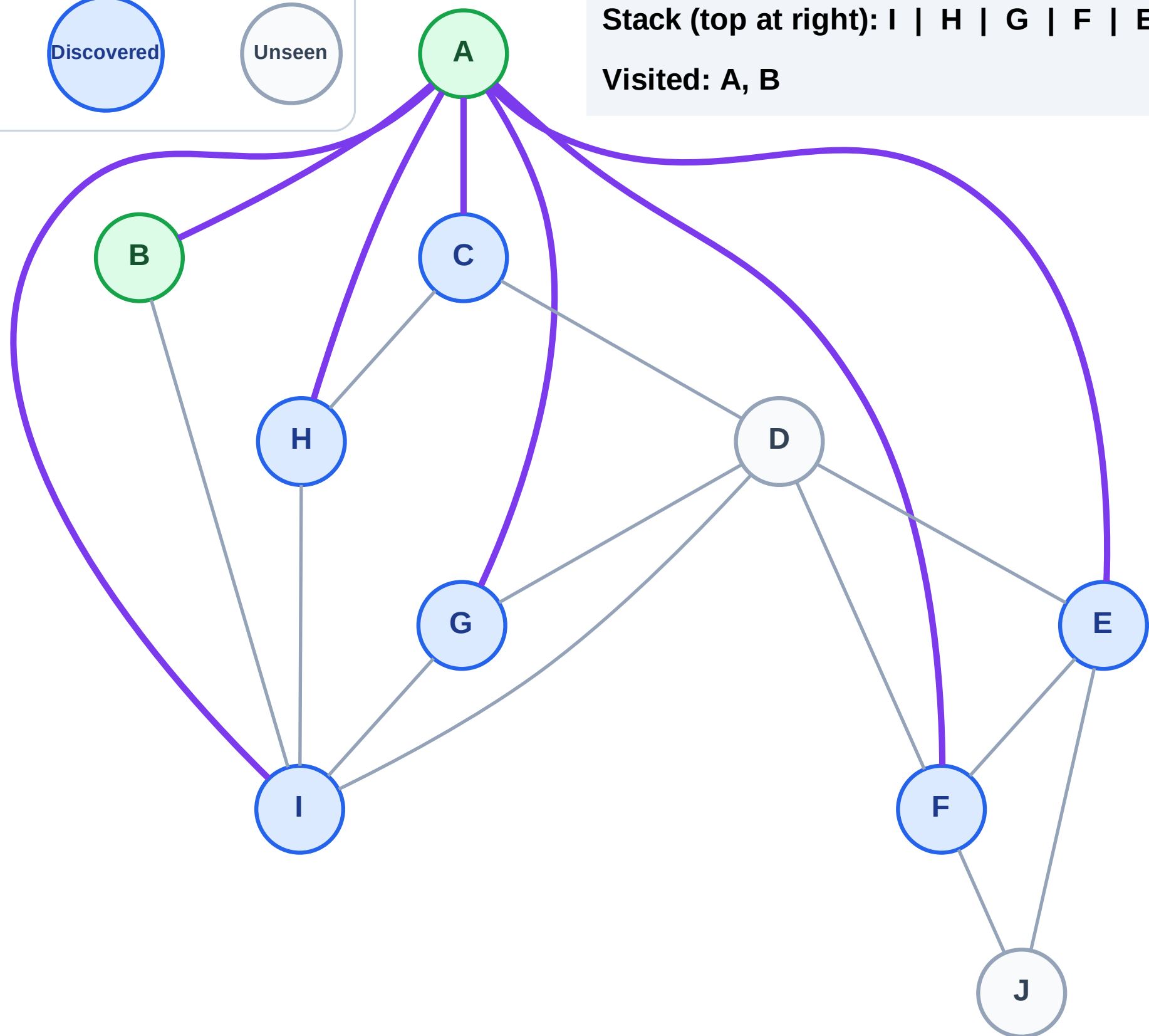
Step 5: No new neighbors from B

Legend



Stack (top at right): I | H | G | F | E | C

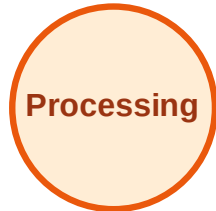
Visited: A, B



DFS Traversal

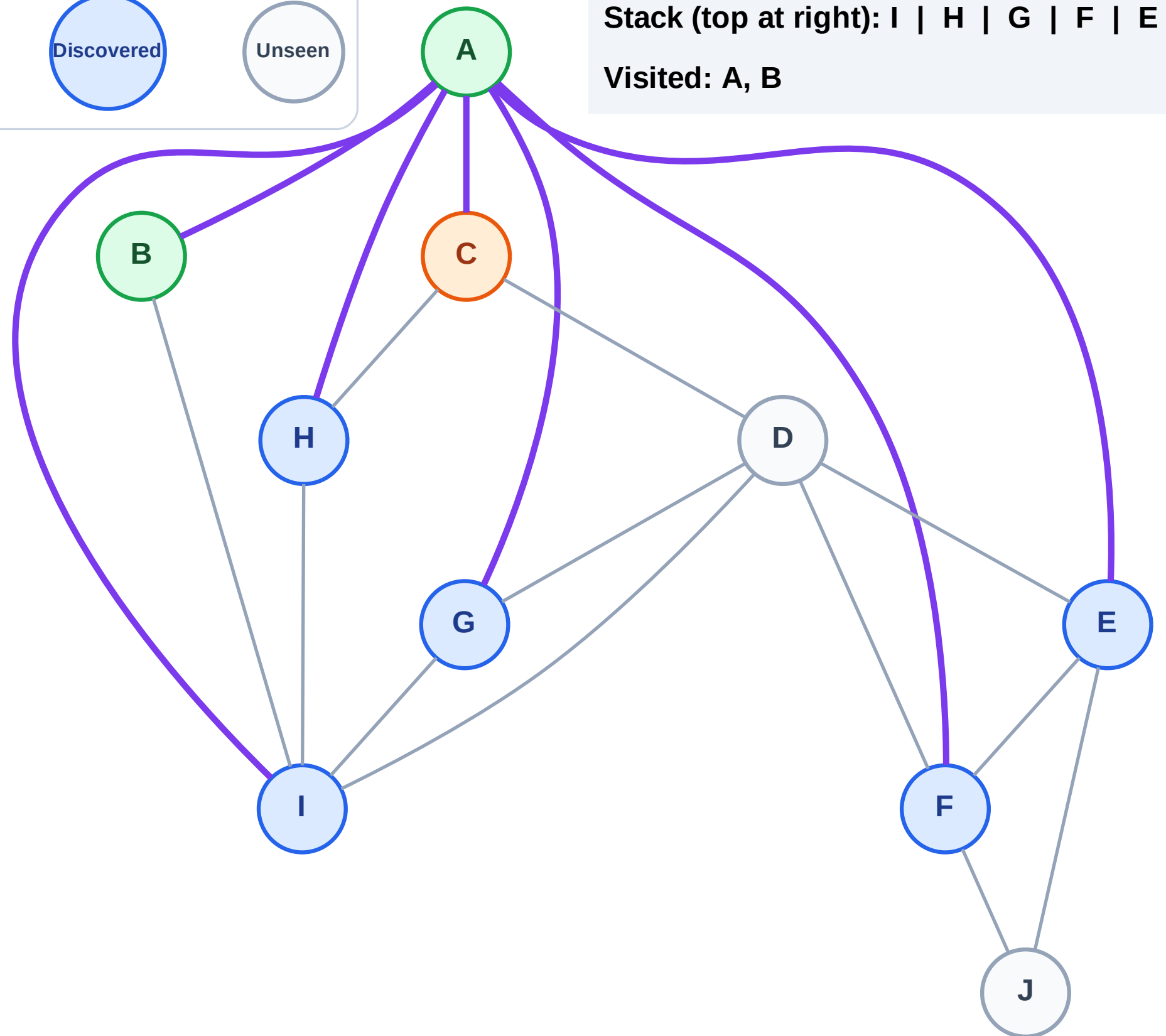
Step 6: Process node C

Legend



Stack (top at right): I | H | G | F | E

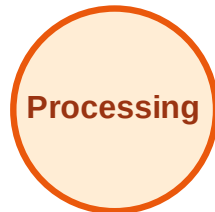
Visited: A, B



DFS Traversal

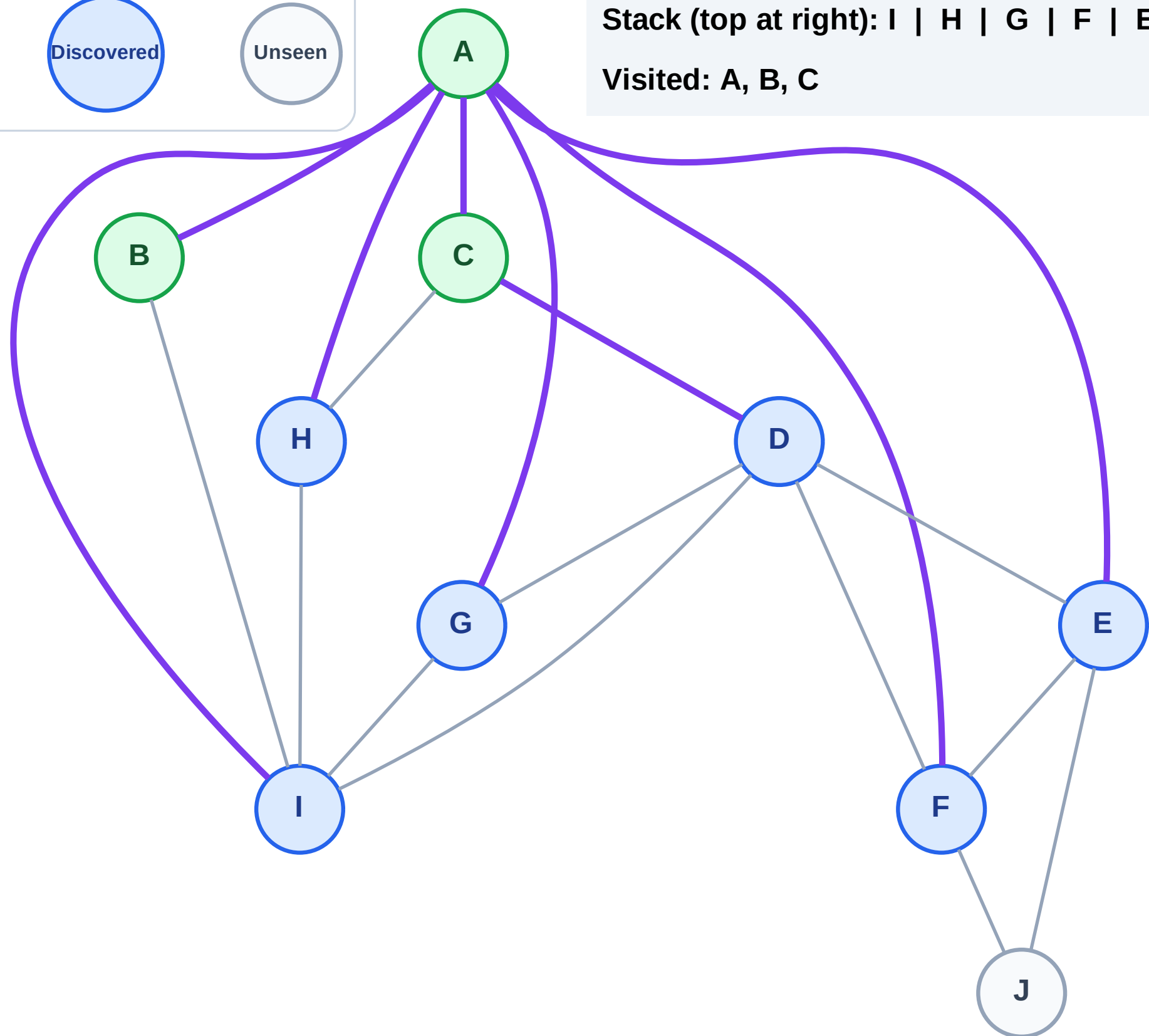
Step 7: Discover D from C

Legend



Stack (top at right): I | H | G | F | E | D

Visited: A, B, C



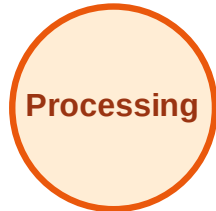
DFS Traversal

Step 8: Process node D

Legend



Complete



Processing



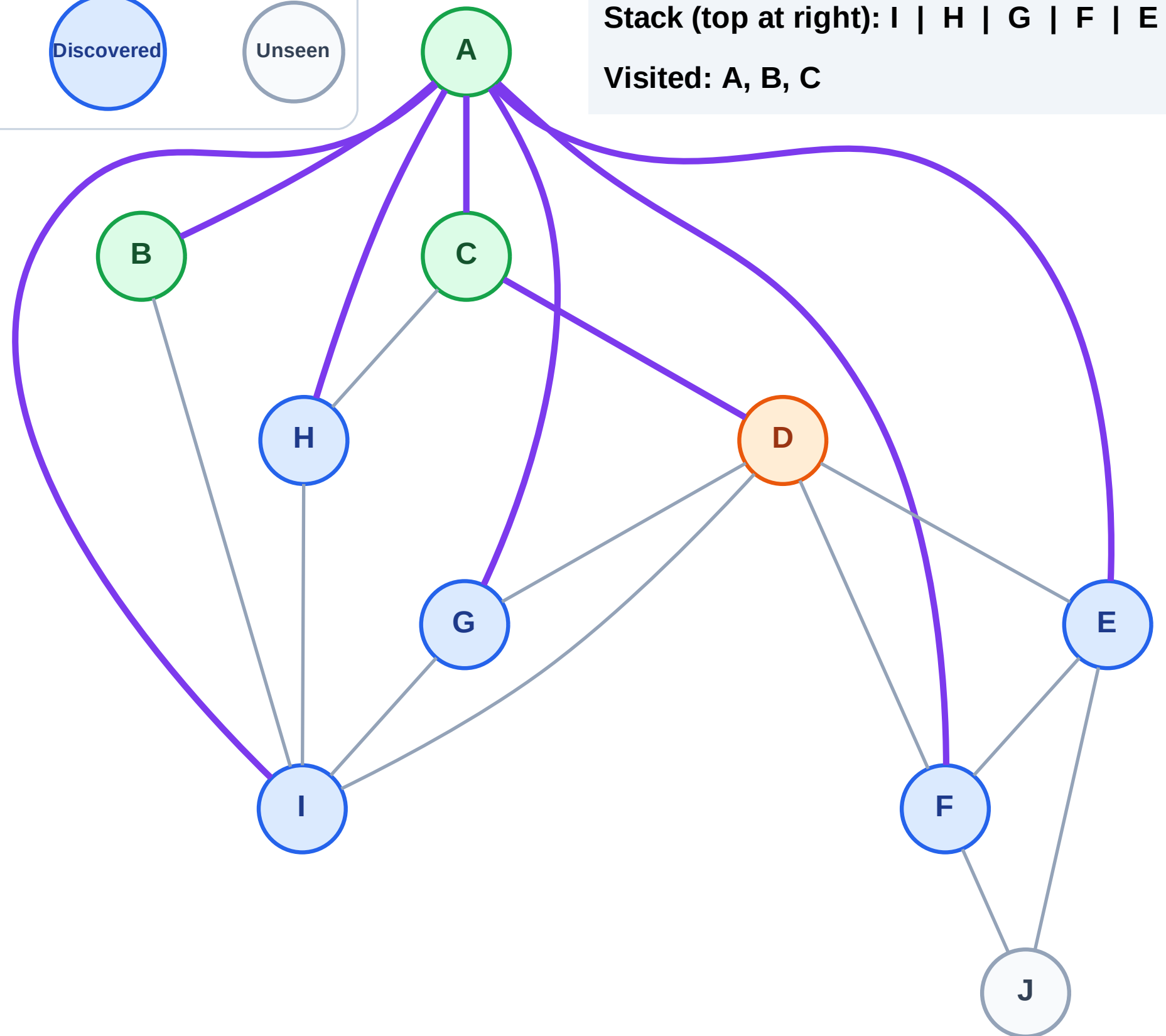
Discovered



Unseen

Stack (top at right): I | H | G | F | E

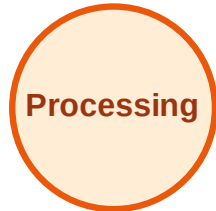
Visited: A, B, C



DFS Traversal

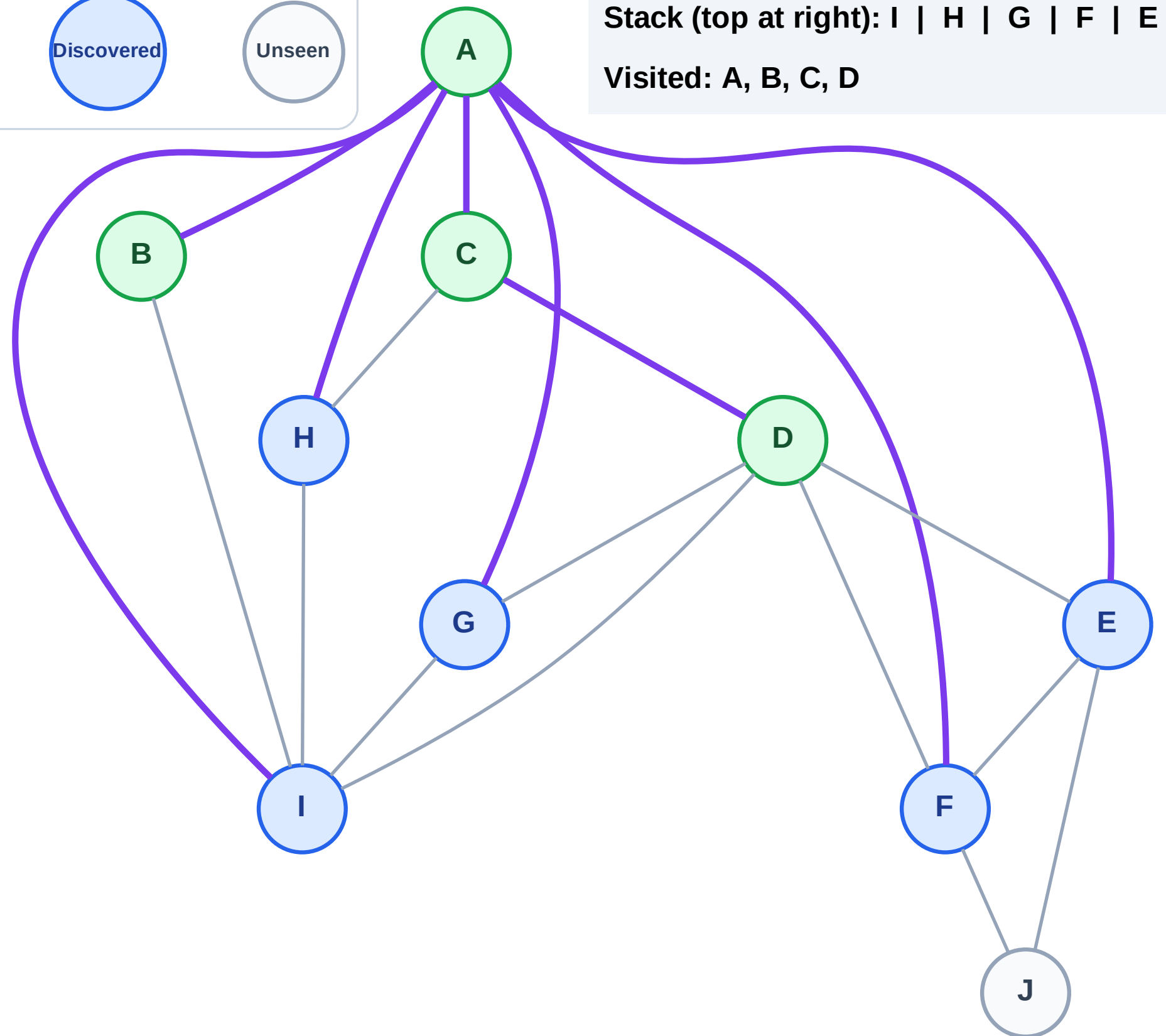
Step 9: No new neighbors from D

Legend



Stack (top at right): I | H | G | F | E

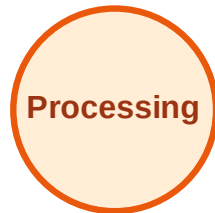
Visited: A, B, C, D



DFS Traversal

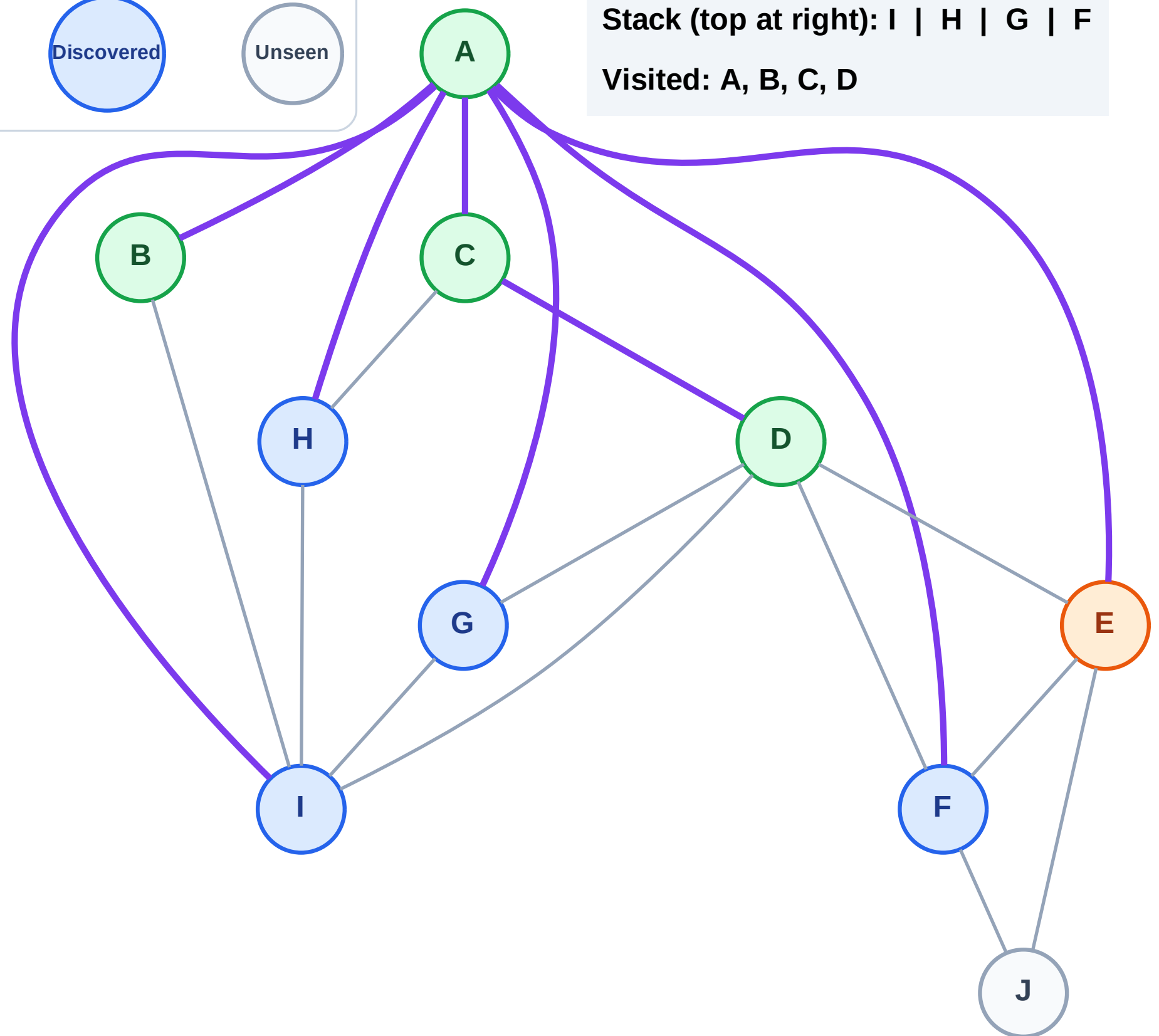
Step 10: Process node E

Legend



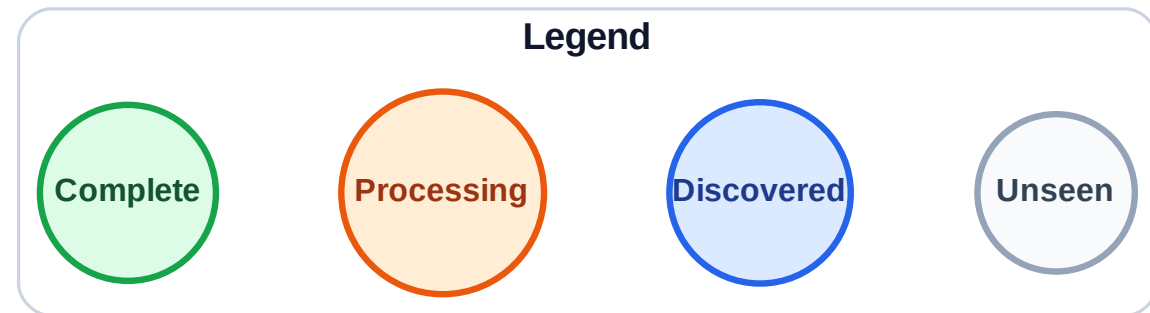
Stack (top at right): I | H | G | F

Visited: A, B, C, D

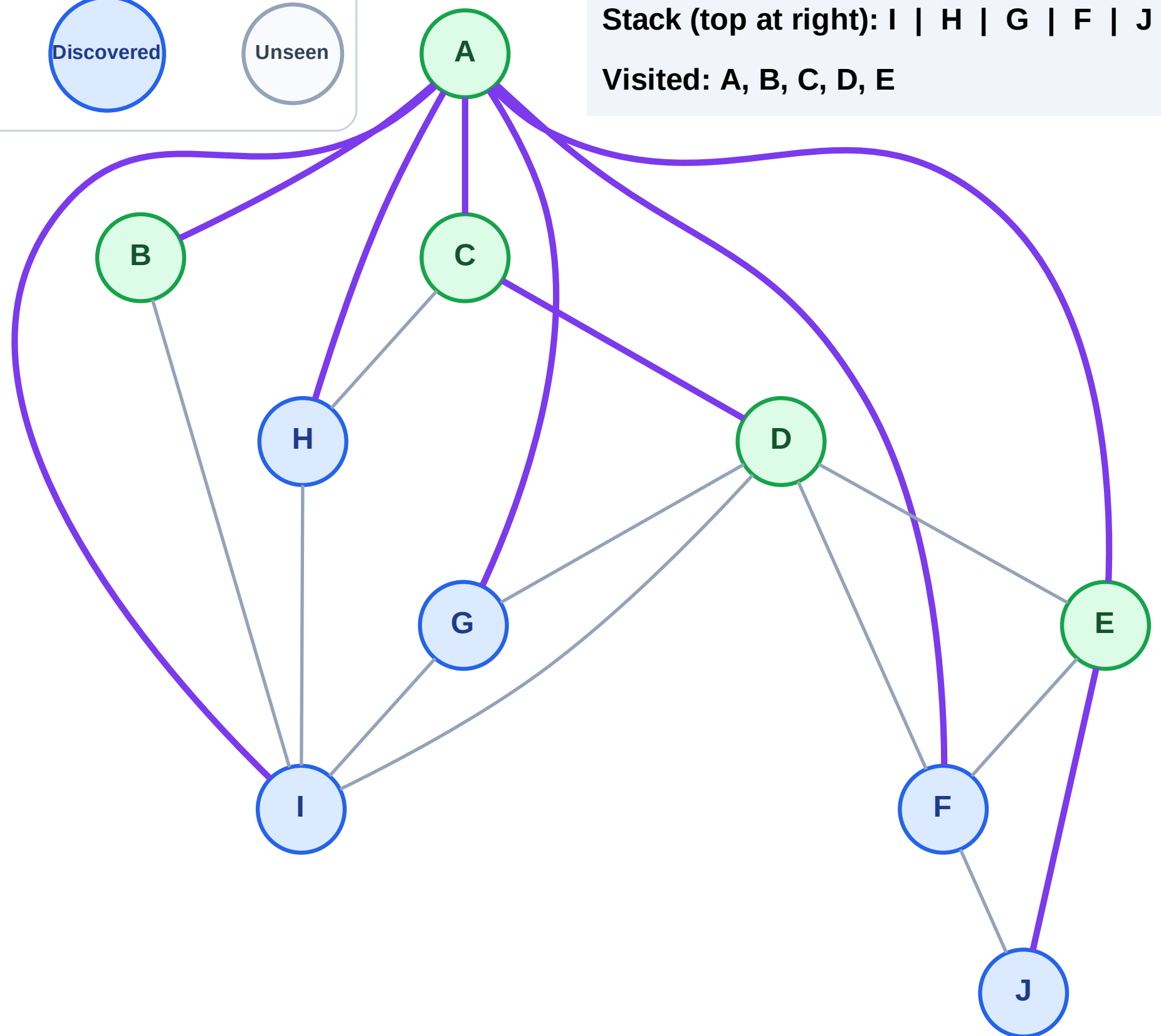


DFS Traversal

Step 11: Discover J from E



Stack (top at right): I | H | G | F | J
Visited: A, B, C, D, E



DFS Traversal

Step 12: Process node J

Legend

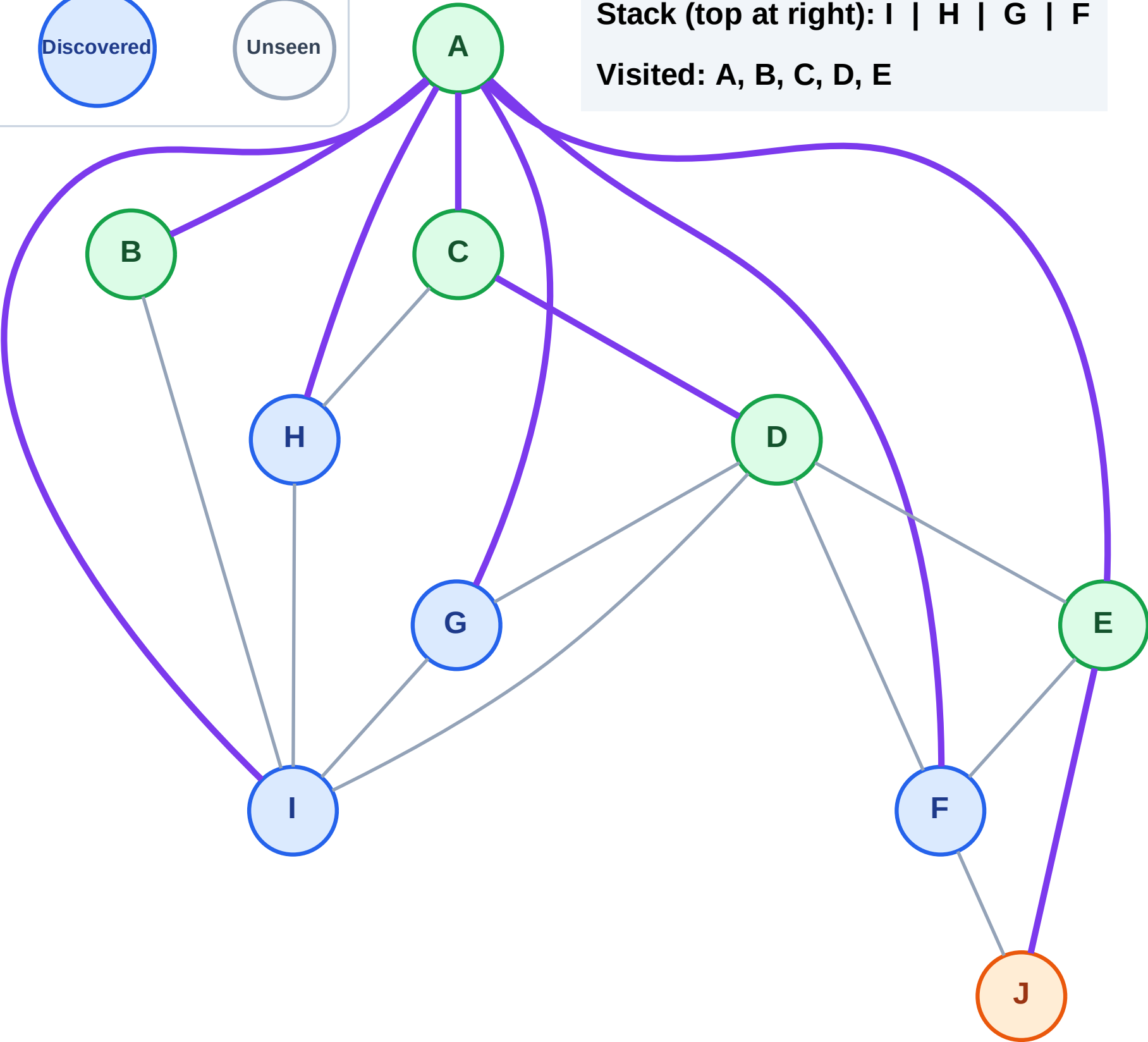
Complete

Processing

Discovered

Unseen

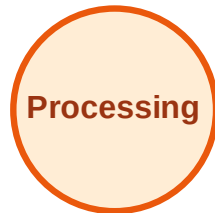
Stack (top at right): I | H | G | F
Visited: A, B, C, D, E



DFS Traversal

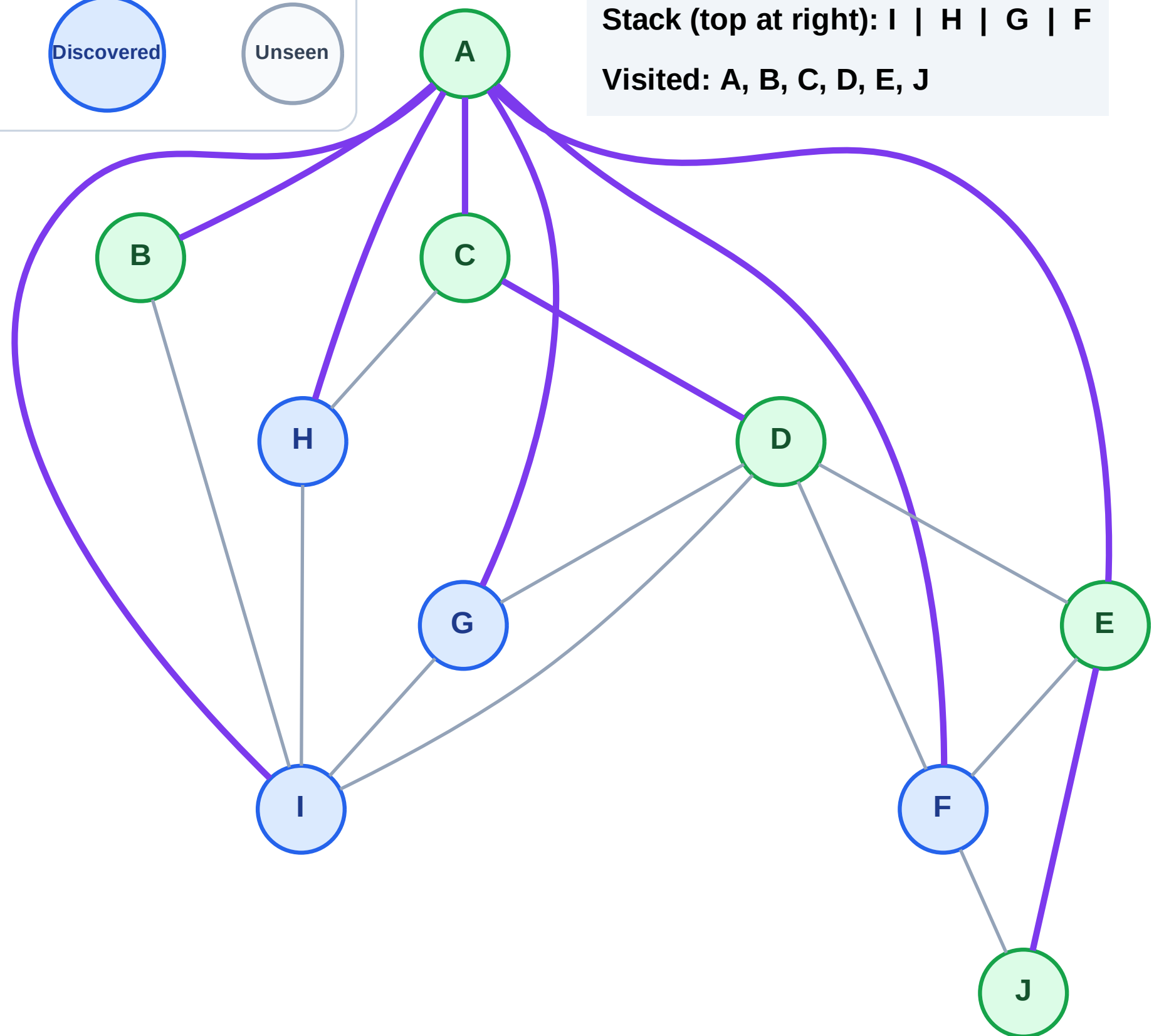
Step 13: No new neighbors from J

Legend



Stack (top at right): I | H | G | F

Visited: A, B, C, D, E, J



DFS Traversal

Step 14: Process node F

Legend

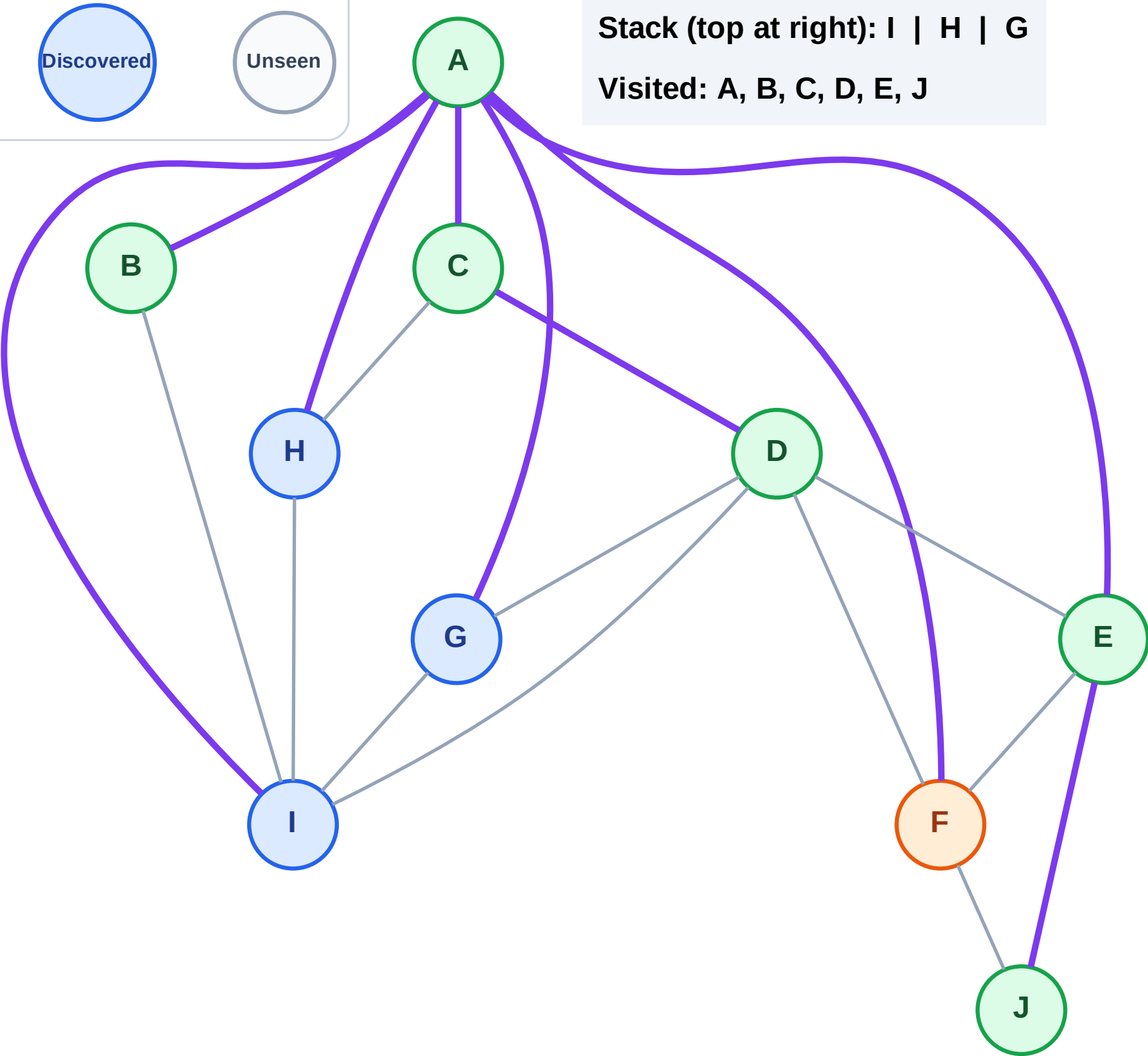
Complete

Processing

Discovered

Unseen

Stack (top at right): I | H | G
Visited: A, B, C, D, E, J

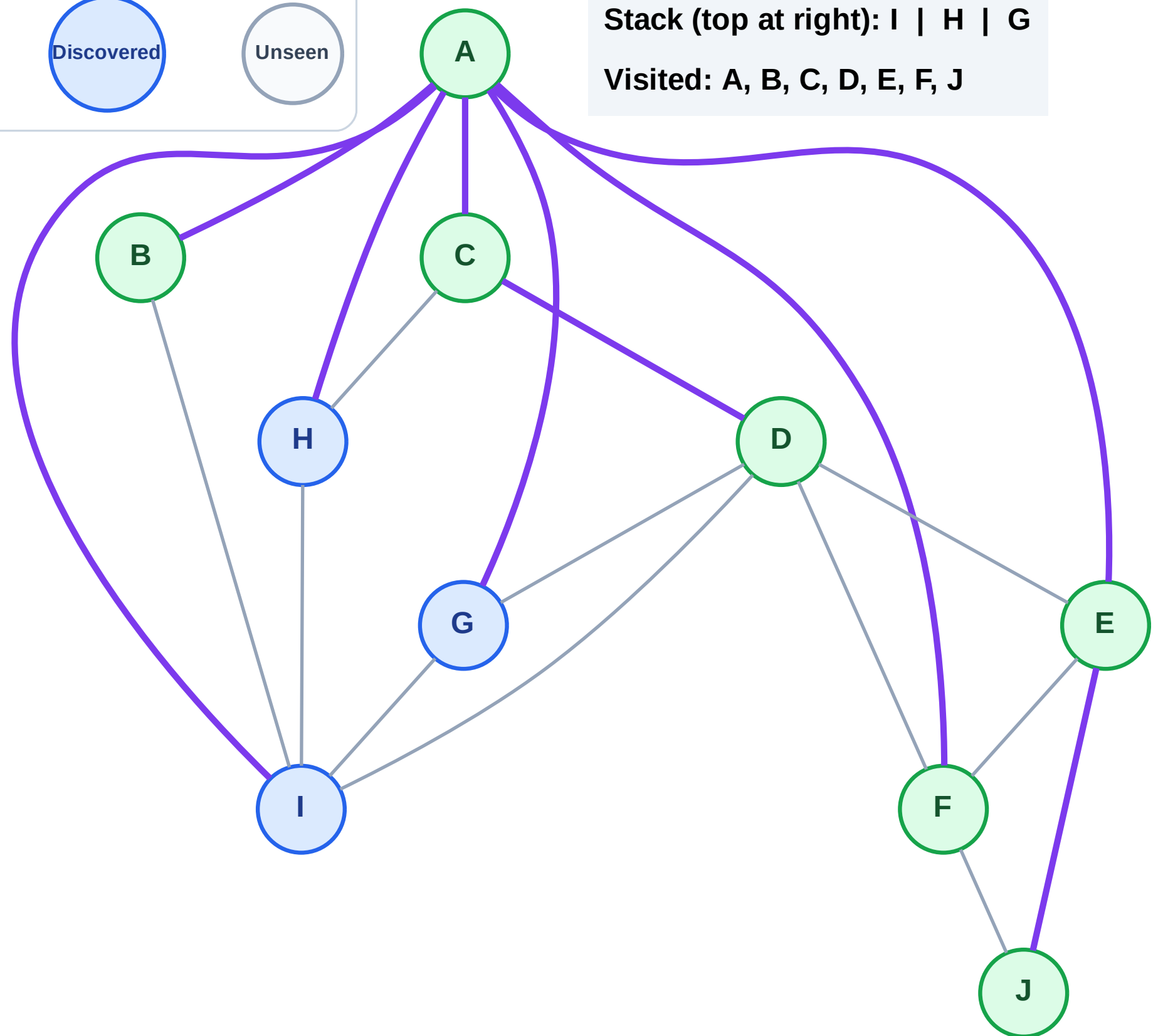


Step 15: No new neighbors from F

Legend

Unseen

Visited: A, B, C, D, E, F, J



DFS Traversal

Step 16: Process node G

Legend

Complete

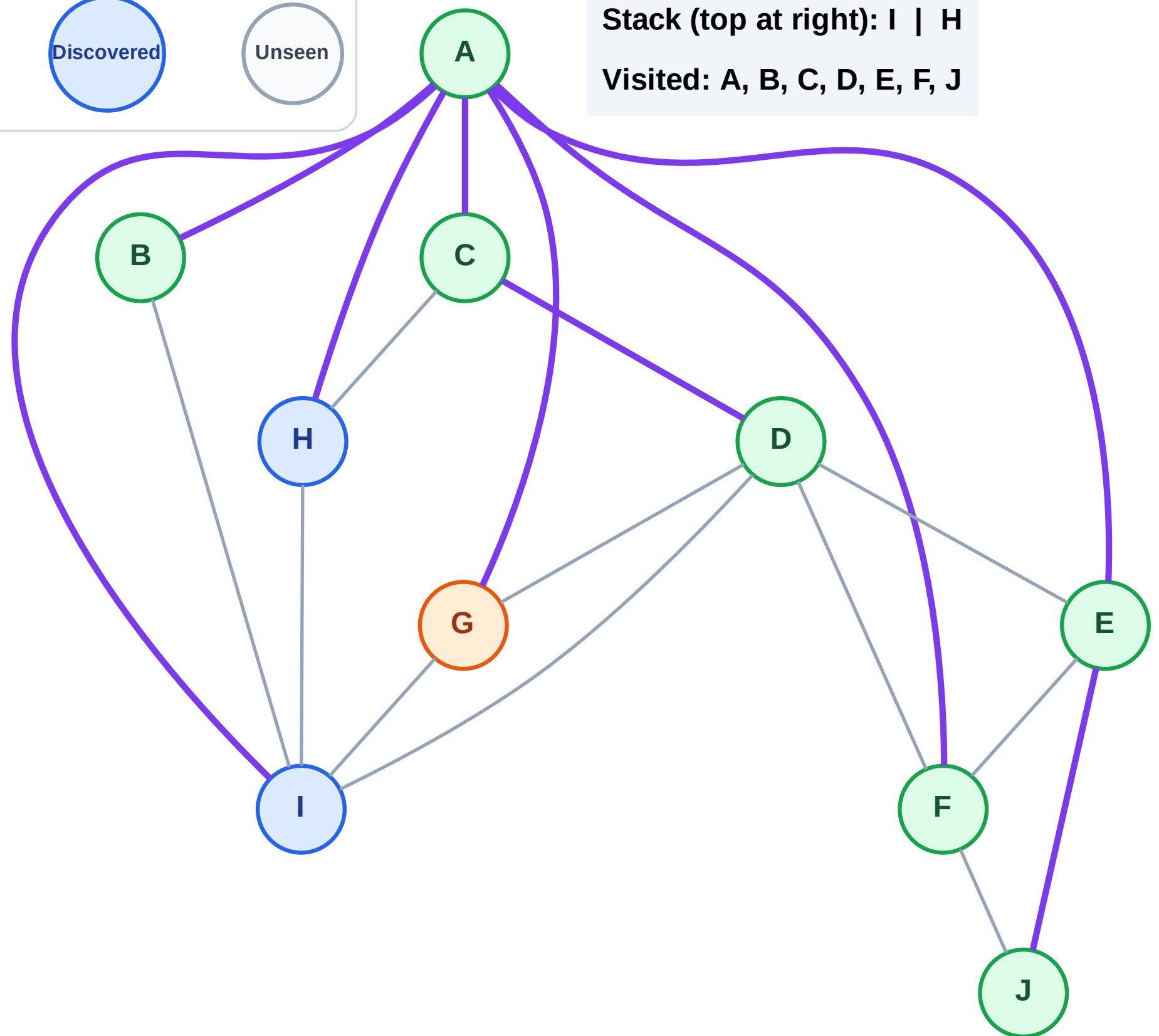
Processing

Discovered

Unseen

Stack (top at right): I | H

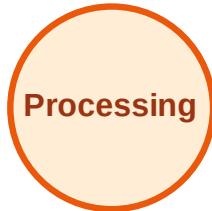
Visited: A, B, C, D, E, F, J



DFS Traversal

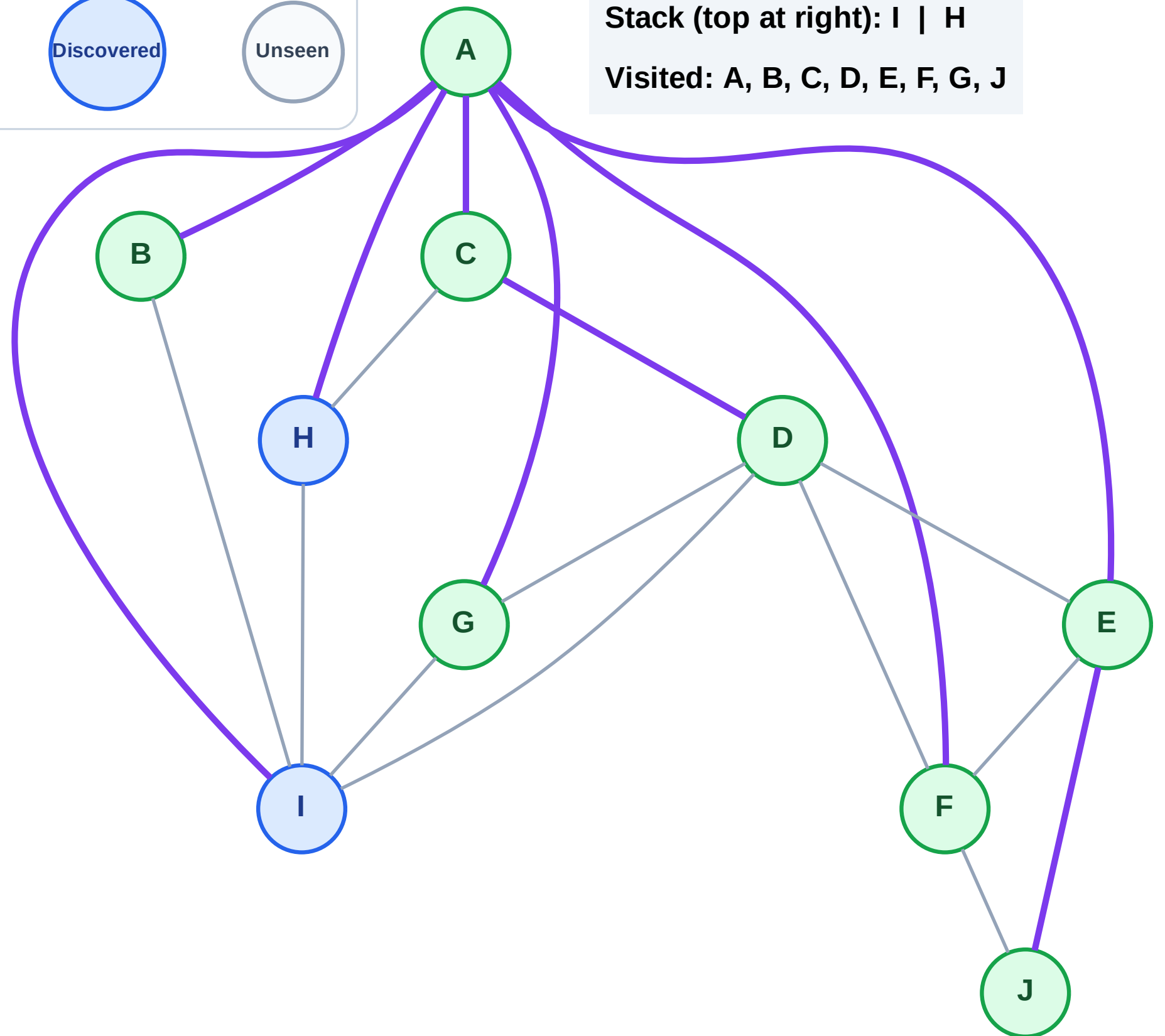
Step 17: No new neighbors from G

Legend



Stack (top at right): I | H

Visited: A, B, C, D, E, F, G, J



DFS Traversal

Step 18: Process node H

Legend

Complete

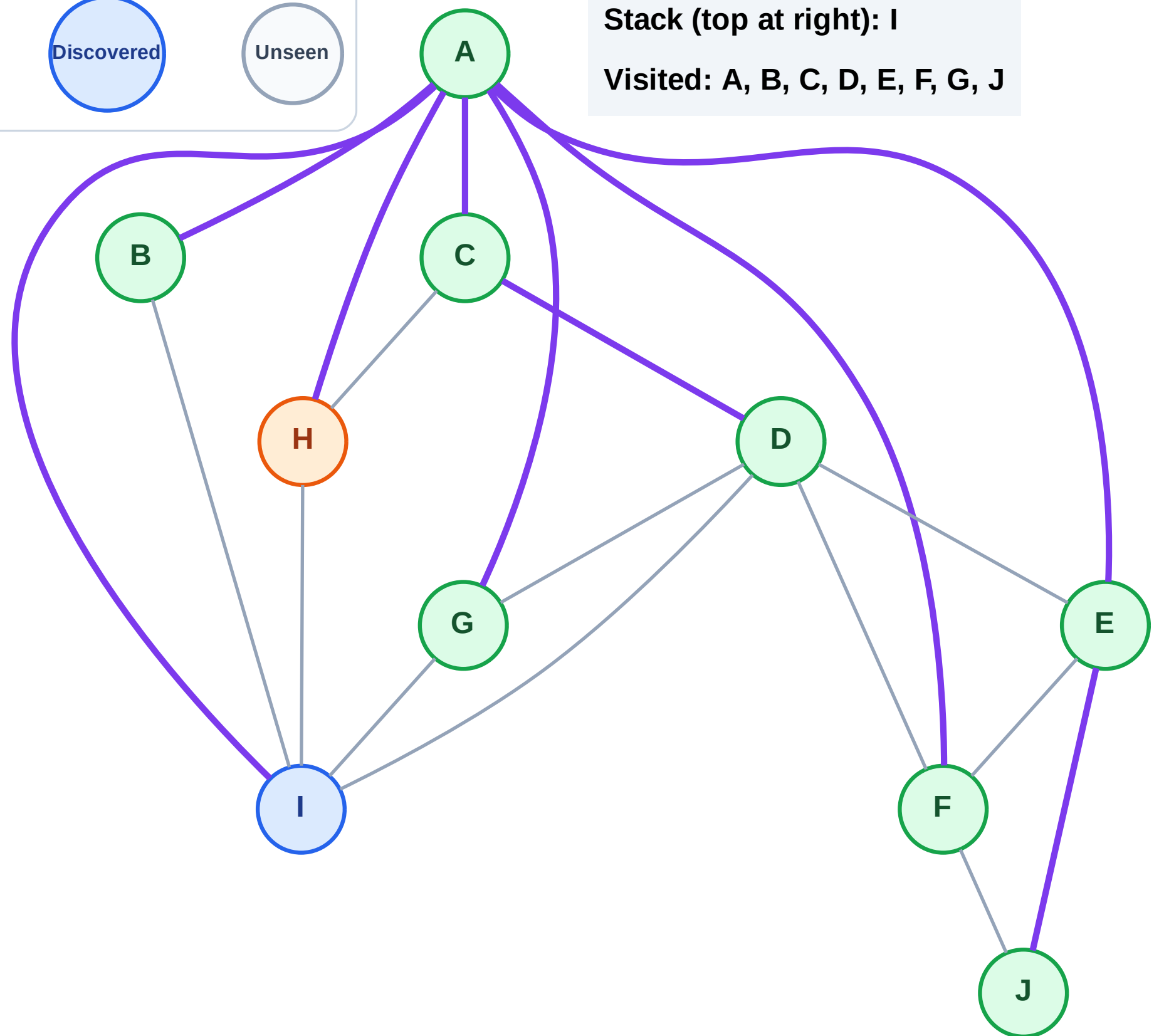
Processing

Discovered

Unseen

Stack (top at right): I

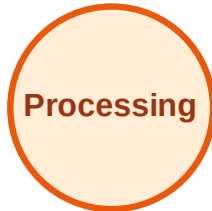
Visited: A, B, C, D, E, F, G, J



DFS Traversal

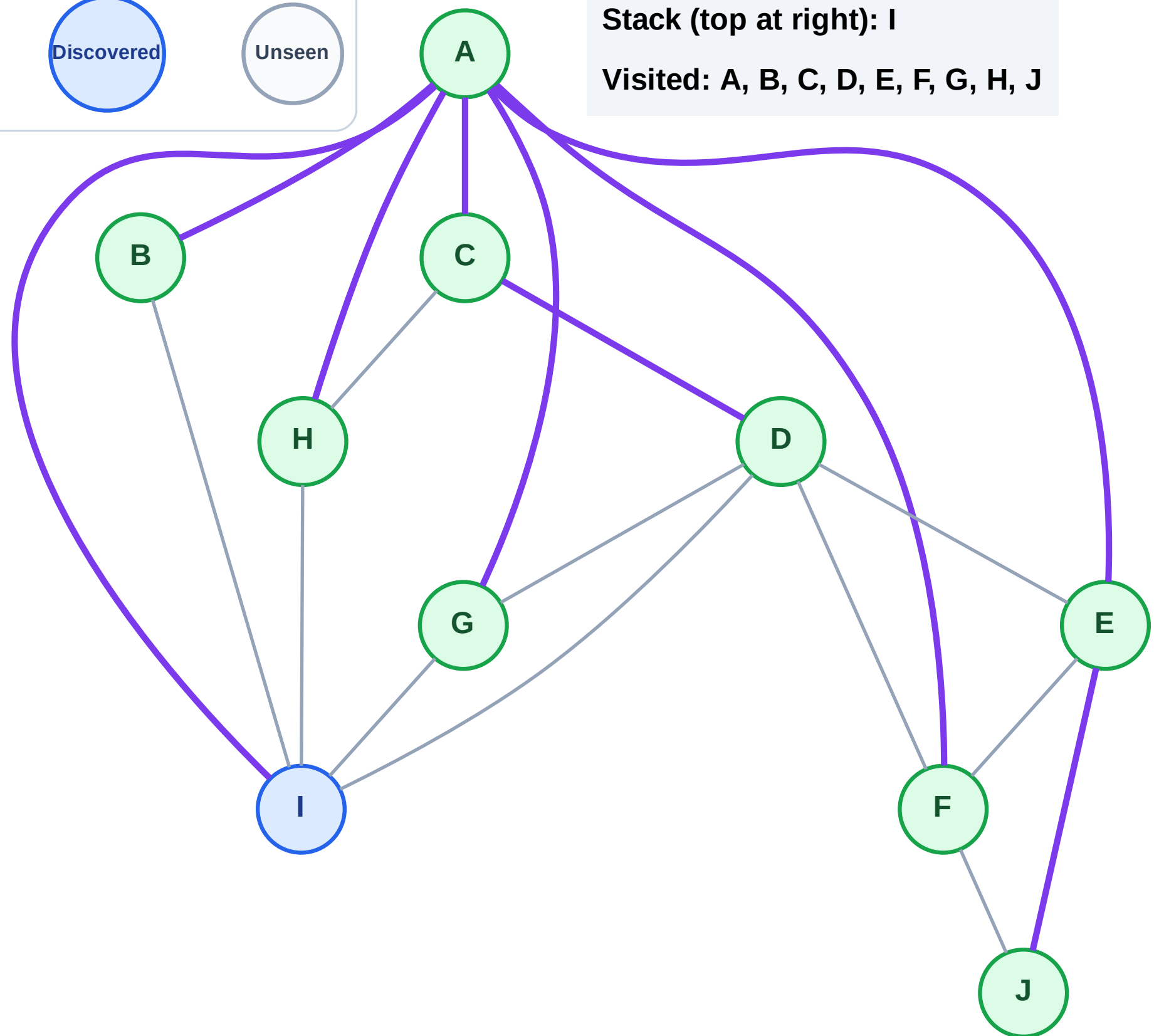
Step 19: No new neighbors from H

Legend



Stack (top at right): I

Visited: A, B, C, D, E, F, G, H, J



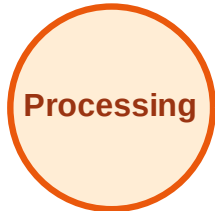
DFS Traversal

Step 20: Process node I

Legend



Complete



Processing



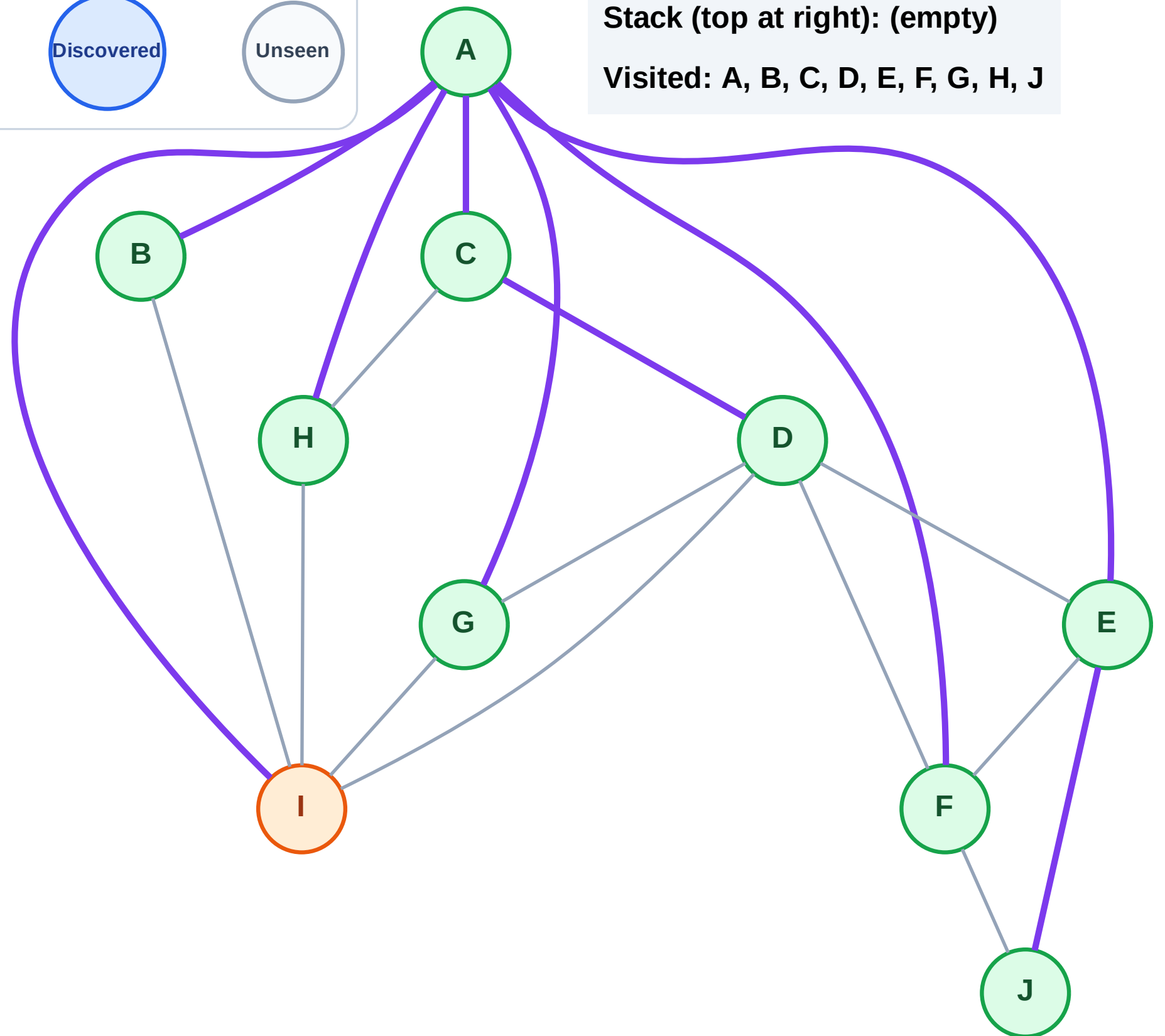
Discovered



Unseen

Stack (top at right): (empty)

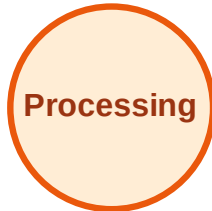
Visited: A, B, C, D, E, F, G, H, J



DFS Traversal

Step 21: No new neighbors from I

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

